CCSDS-SAT-TTC-IP (v2.40.3)
Product Guide & Hardware Integration Manual

Vendor-Agnostic IEEE 1800-2017 SystemVerilog Transceiver IP Core, Pinout & Register Reference, Bare-Metal ISO C11 HAL SDK, and S-Band Board Integration & Verification Guide

Document Attribute	Specification Summary
Document ID & Companion	PG-CCSDS-TTC-v2.40.3 (Volume I) — Companion to **Volume II: Engineering Architecture & Mathematical Monograph** (ENG-CCSDS-TTC-v2.40.3)
Top-Level RTL Entity	ccsds_sat_ttc_top (rtl/ccsds_sat_ttc_top.sv, 30 synthesizable/model SV files)
Hardware Version ID	32'hCC5D_0230 (CORE_ID_VER register at AXI4-Lite offset 0x00)
Synthesizable Personas	ROLE_MODE = 0 (Spacecraft Flight, 23 modules), ROLE_MODE = 1 (Ground EGSE, 22 modules), ROLE_MODE = 2 (Symmetric Crosslink & On-Orbit BIST, 28 modules)
Standards Compliance	17 CCSDS Blue & Orange Books: Space Packet (133.0-B-2) [1], TM (132.0-B-3) [2], TC (232.0-B-4) [3], COP-1 (232.1-B-2) [4], AOS (732.0-B-4) [5], USLP (732.1-B-3) [6], Proximity-1 (211.0/1/2-B) [7], TM Sync/Coding (131.0-B-5) [8], TC Sync/Coding (231.0-B-4) [9], Short-Block Uplink LDPC (231.1-O-1) [10], CUC OBT (301.0-B-4) [11], SDLS (355.0/1-B) [12], RF/Modulation (401.0-B-32) [13], PN Ranging (414.1-B-3) [14]
Hardware Independence	**100% Vendor-Agnostic RTL (0 BRAM / 0 DSP)**: Uses inferred Distributed LUTRAM (RAM64M) + Flip-Flops; synthesizes unmodified across AMD/Xilinx, Lattice, Microchip, and NanoXplore FPGAs
Static Timing Closure	110.22–169.46 MHz Zynq-7010 (-1L); 203.38–271.74 MHz Kintex UltraScale True $F_{\max}$ across all 4 clock domains
Flight Software SDK	Zero-dependency Bare-Metal ISO C11 HAL (sw/ccsds_ttc_hal.h, sw/ccsds_ttc_hal.c)
Verification Sign-Off	100.0% Pass across all **5,760 valid CCSDS RF configurations** × **6 channel profiles (34,560 trials)** with **0 post-FEC bit errors** at rated mission SNR in Xilinx Vivado 2022.2 xsim

Role Separation Between Volume I and Volume II: **Volume I (this Product Guide)** is the practical hardware integration and software programming manual: it covers the 30-module file inventory, compile-time generics, 4-clock timing closure, 59-port signal dictionary, complete 0x00..0x50 register map, function-by-function C11 HAL SDK reference, S-Band board LVDS/SPI/protection wiring & bring-up guide, and regression test execution. **Volume II (ENG-CCSDS-TTC-v2.40.3)** is the companion theoretical monograph containing all first-principles finite-field proofs, DSP/FEC/cryptographic mathematical derivations, Friis noise-temperature cascades, and orbital link physics.

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1 Architecture Overview, RTL Inventory & Signal Flow

Every space mission—from a Low-Earth Orbit (LEO) CubeSat to a geostationary relay or lunar lander—depends on its Telemetry, Tracking, and Command (TT&C) radio link as its ultimate survival lifeline. This chapter introduces the vendor-agnostic design philosophy of `CCSDS-SAT-TTC-IP` (`ccsds_sat_ttc_top`), explains its three compile-time personas, walks stage-by-stage through the uplink and downlink signal pipelines, and catalogs all 30 SystemVerilog modules and 17 supported CCSDS standards.

1.1 Vendor-Agnostic Design & Operating Personas

The **`CCSDS-SAT-TTC-IP` (`ccsds_sat_ttc_top`)** is a radiation-mitigated digital baseband modem, Forward Error Correction (FEC) codec, multi-protocol data link framer/deframer, closed-loop Automatic Repeat Request (ARQ) controller, and inline cryptographic security core written in synthesizable IEEE 1800-2017 SystemVerilog. The core instantiates zero proprietary FPGA hard macros (**0 Block RAMs** and **0 DSP slices**) and exposes standard signed two's-complement In-Phase and Quadrature (`I/Q`) sample buses (`rx_i_adc_in`, `rx_q_adc_in`, `tx_i_dac_out`, `tx_q_dac_out`) that interface cleanly to any external ADC/DAC or zero-IF SDR transceiver.

To eliminate flight-to-ground implementation asymmetry while pruning unused logic in orbit, the top-level parameter `ROLE_MODE` specializes the hierarchy into one of three synthesizable personas sharing **78.6% of the RTL codebase** (**Figure 1.1**):

1. **Spacecraft Flight Transceiver (`ROLE_MODE = 0` — 23 Modules):** Elaborates the 17 shared core modules plus the 6 spacecraft endpoint modules: uplink BCH$(63, 56)$ and Short-LDPC$(128, 64)$ decoders, 3x Triple-Modular Redundant (TMR) `FARM-1` responder, 3x TMR Emergency Hardware Command Pulse Decoder (`HCPD`), and downlink RS$(255, 223)$ and Turbo/AR4JA encoders.[1]
2. **Hardware Ground EGSE (`ROLE_MODE = 1` — 22 Modules):** Elaborates the 17 shared core modules plus the 5 ground-station endpoint modules: `FOP-1` Go-Back-N command initiator (`ccsds_fop1_engine.sv`), TC/-CLTU framer (`ccsds_tc_cltu_framer.sv`), ground BCH$(63, 56)$ encoder (`ccsds_bch6356_enc.sv`), $t = 16$ Berlekamp-Massey RS$(255, 223)$ decoder (`ccsds_dl_rs255_223_dec.sv`), and ground TM/AOS deframer (`ccsds_tm_aos_deframer.sv`).
3. **Universal Crosslink & On-Orbit BIST (`ROLE_MODE = 2` — All 28 Modules):** Elaborates both flight and ground endpoints simultaneously for symmetric Proximity-1/USLP inter-satellite crosslinks and autonomous on-orbit loopback Built-In Self-Test (BIST).

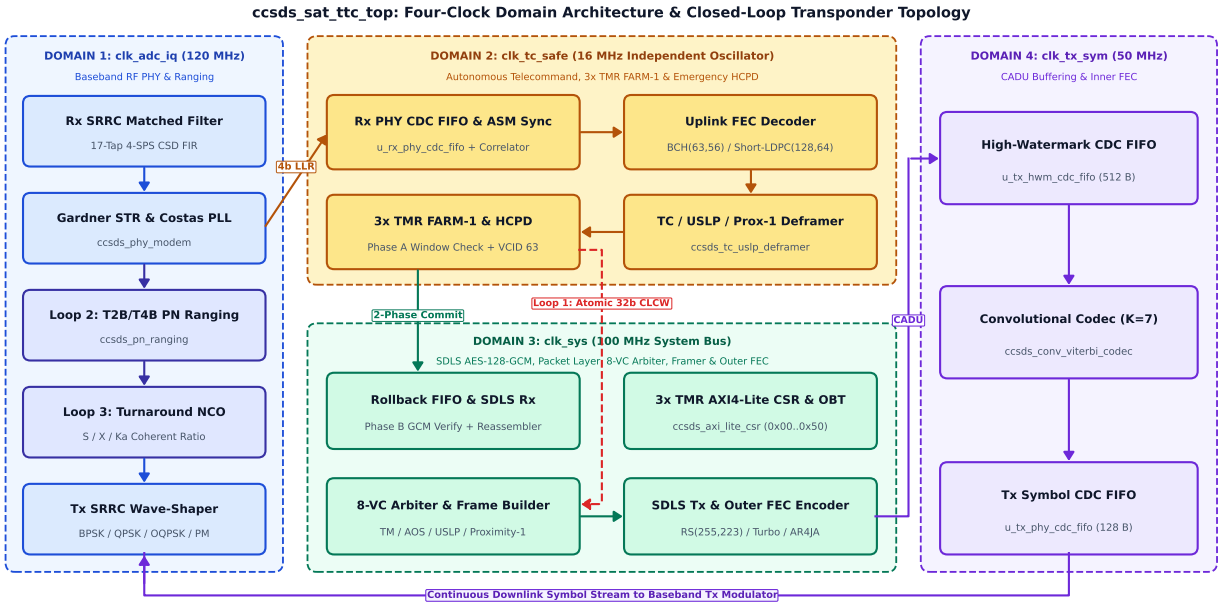


Figure 1.1: Four-clock domain architecture and closed-loop transponder topology of `ccsds_sat_ttc_top`.

[1] Ground-only endpoint blocks—specifically the $t = 16$ Berlekamp-Massey RS$(255, 223)$ decoder, the 64-state soft Viterbi decoder in `ccsds_conv_viterbi_codec.sv`, and the `FOP-1` retransmission queue—are pruned during synthesis when `ROLE_MODE = 0`.

1.2 Seven-Stage Uplink & Downlink Signal Flow

Figure 1.2 illustrates the end-to-end signal processing pipelines for an incoming uplink telecommand (**Stages U1–U7**) and an outgoing downlink telemetry frame (**Stages D1–D7**), together with the three closed-loop transponder paths connecting receiver to transmitter.

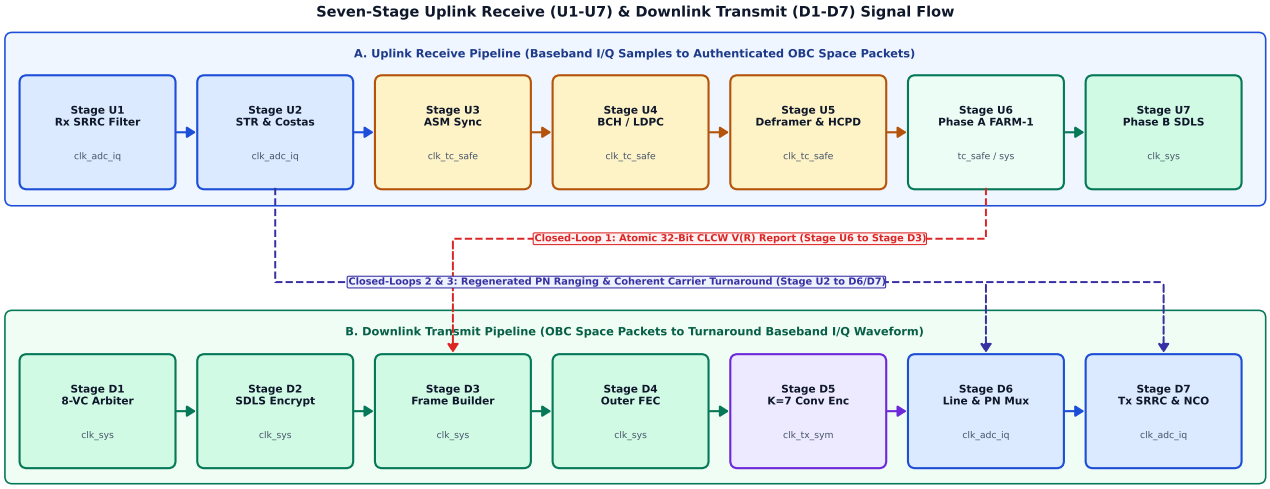


Figure 1.2: Seven-stage uplink receive pipeline (U1–U7), downlink transmit pipeline (D1–D7), and closed-loop transponder feedback paths.

A. Uplink Receive Pipeline (Stages U1–U7):

1. **Stage U1 — Baseband Filtering (clk_adc_iq, 120 MHz):** Complex NCO derotation and the 17-tap multiplier-free CSD 4-SPS SRRC matched filter ($\alpha = 0.35$) in ccsds_phy_modem.sv strip out-of-band noise and close the Nyquist eye.
2. **Stage U2 — Carrier, Timing & Ranging Lock (clk_adc_iq):** An FLL-assisted PI Costas loop tracks carrier Doppler, a 3-stage Gardner TED slices 4-bit soft LLRs, and ccsds_pn_ranging.sv locks onto the T2B/T4B ranging sequence.
3. **Stage U3 — CDC Crossing & Frame Sync (clk_tc_safe, 16 MHz):** Soft LLRs cross via u_rx_phy_cdc_fifo into the independent telecommand clock domain, where ccsds_asm_correlator.sv resolves $0°/180°$ phase ambiguity.
4. **Stage U4 — Uplink FEC & Descrambling (clk_tc_safe):** Hardware straps (ul_fec_override_en) select expurgated BCH$(63, 56)$ SEC-DED or 4-phase Short-LDPC$(128, 64)$ Soft-OMS decoding, followed by $h_{\mathrm{TC}}(x)$ descrambling.
5. **Stage U5 — Deframing & Emergency HCPD (clk_tc_safe):** ccsds_tc_uslp_deframer.sv strips 0x55 fill octets and checks CRC-16/32, while VCID=63 frames branch directly to 3x TMR ccsds_hcpd_ctrl.sv for OBC hardware reset.
6. **Stage U6 — Phase A Window Check (clk_tc_safe → clk_sys):** 3x TMR ccsds_farm1_tmr.sv verifies $N(S)$ against the sliding window while keeping $V(R)$ frozen and writes the candidate frame into u_rx_commit_cdc_fifo.
7. **Stage U7 — Phase B SDLS Verify & Packet Release (clk_sys, 100 MHz):** ccsds_sdls_engine.sv verifies the 128-bit AES-128-GCM tag; a valid tag commits $V(R)$ (**Closed-Loop #1**) and releases packets on m_axis_rx_pkt_*, whereas a failed tag rolls back the FIFO pointer.

B. Downlink Transmit Pipeline (Stages D1–D7):

1. **Stage D1 — Virtual Channel Arbitration (clk_sys, 100 MHz):** ccsds_vc_arbiter_mux.sv arbitrates across 8 VCs from s_axis_tx_pkt_* and computes the Cycle-0 First_Hdr_Ptr (FHP).
2. **Stage D2 — Inline SDLS Encryption (clk_sys):** ccsds_sdls_engine.sv encrypts the Transfer Frame Data Field using AES-128-CTR/GCM and appends the 128-bit GHASH authentication tag.

3. **Stage D3 — Multi-Protocol Frame Assembly (clk_sys):** ccsds_tm_aos_uslp_framer.sv attaches the Primary Header, 48-bit CUC OBT, live 32-bit CLCW (**Closed-Loop #1**), and CRC-16/32 trailer.
4. **Stage D4 — Outer FEC & Randomization (clk_sys):** ccsds_dl_rs255_223_enc.sv (or ccsds_dl_turbo_ldpc_enc.sv) and the $h_{\text{TM}}(x)$ scrambler form the Channel Access Data Unit (CADU).
5. **Stage D5 — High-Watermark Buffering & Inner FEC (clk_tx_sym, 50 MHz):** The 512-byte pre-fill FIFO u_tx_hwm_cdc_fifo feeds ccsds_conv_viterbi_codec.sv ($r = 1/2, K = 7$) with zero underflow bubbles.
6. **Stage D6 — Line Coding & Ranging Overlay (clk_adc_iq, 120 MHz):** Symbols map to NRZ-L/M or Bi-ϕ-L over BPSK/QPSK/OQPSK/PCM-PM with regenerated T2B/T4B ranging overlaid on the Q channel (**Closed-Loop #2**).
7. **Stage D7 — Polyphase SRRC Shaping & Turnaround (clk_adc_iq):** The 17-tap polyphase SRRC filter and rational turnaround NCO (**Closed-Loop #3**: S-Band $240/221$, X-Band $880/749$, or X→Ka $3344/749, 3360/749, 3328/749$) drive tx_i/q_dac_out.

1.3 RTL Module Inventory & Standards Compliance

Table 1.1 organizes all 30 SystemVerilog files in rtl/ and the 11 testbenches in tb/ into five functional groups.

Together, the synthesizable modules in **Table 1.1** implement 17 CCSDS Blue and Orange Book standards across packet formatting, data link framing, forward error correction, link security, timekeeping, and physical-layer modulation. **Table 1.2** maps each CCSDS specification to its corresponding RTL implementation.

2 Product Specification, Clocking & Resource Footprint

Having traced the functional signal flow across the transceiver in Chapter 1, we now specify the physical FPGA integration parameters required to instantiate ccsds_sat_ttc_top. This chapter details each compile-time synthesis generic and its legal values, explains why the architecture is partitioned across four isolated clock domains, defines the reset and 1-PPS timing requirements, and reports measured Logic LUT, Distributed LUTRAM (RAM64M), and Flip-Flop resource utilization and static timing closure.

2.1 Compile-Time Synthesis Generics

To allow avionics designers to trade off optional protocol features against FPGA LUT footprint without editing RTL source files, ccsds_sat_ttc_top exposes four compile-time SystemVerilog parameters. In **Table 2.1**, every legal parameter value is given its own dedicated row to clearly show which sub-modules are elaborated or pruned and how FPGA logic resources are affected.

2.2 Four-Clock Architecture & CDC Isolation

In a flight-qualified spacecraft transponder, clocking the emergency command receiver from the same Phase-Locked Loop (PLL) as the On-Board Computer bus creates a single-point failure: if an OBC clock synthesizer unlocks or stalls, the satellite becomes uncommandable. To eliminate this hazard and match each pipeline stage to its natural physical rate, ccsds_sat_ttc_top partitions its logic across four asynchronous clock domains (**Table 2.2**) joined by Gray-coded Hsiao $(39, 32)$ SECDED Distributed-LUTRAM FIFOs and 4-phase Request/Acknowledge handshakes.

2.3 Reset Sequencing & 1-PPS Time Synchronization

Proper initialization across all four asynchronous clock domains is handled internally by dedicated reset synchronizers, while spacecraft timekeeping is maintained by a hardware 1-PPS synchronization circuit:

- **Per-Domain 2-FF Reset Bridges (rst_n):** The single active-low master reset pin rst_n feeds four independent 2-flip-flop asynchronous-assert, synchronous-deassert reset synchronizer bridges (rst_sys_n, rst_tc_n, rst_tx_n, rst_adc_n marked with (* ASYNC_REG = "TRUE" *)). During power-up or watchdog reset, hold rst_n == 1'b0 for at least **8 cycles of the slowest clock (clk_tc_safe, ≥ 500 ns at 16 MHz)** to guarantee clean, recovery/removal-safe deassertion across all four domains.

SystemVerilog File Path	Clock Domain	ROLE_MODE	Primary Hardware Function
1. Shared Foundation, Math & Clock-Domain Crossing Primitives (6 Files)			
rtl/ccsds_pkg.sv	All Domains	0, 1, 2	Packed protocol structs, $GF(2^8)$ dual-basis matrices, and FEC constants.
rtl/ccsds_crc_universal.sv	clk_sys / tc	0, 1, 2	Single-cycle byte-wide CRC-16-CCITT (0x1021) and CRC-32 (0x00A00805).
rtl/ccsds_lfsr_scrambler.sv	clk_sys / tc	0, 1, 2	255-byte $h_{\text{TM}}(x)$ (0xA9) and $h_{\text{TC}}(x)$ (0x5F) additive randomizers.
rtl/ccsds_cdc_fifo_ecc.sv	Dual-Clock	0, 1, 2	Gray-coded LUTRAM (RAM64M) CDC FIFO with Hsiao $(39, 32)$ SECDED & rollback.
rtl/ccsds_gf256_math.sv	clk_sys	0, 1, 2	$GF(2^8)$ multiplier (0x187), dual-basis map, and AOS $GF(2^4)$ RS$(10, 6)$ FHEC.
rtl/ccsds_axi_lite_csr.sv	clk_sys	0, 1, 2	3x TMR AXI4-Lite CSR bank (0x00..0x50), 48-bit CUC OBT, and W1C counters.
2. Uplink Telecommand, Security, COP-1 ARQ & Emergency Recovery Chain (9 Files)			
rtl/ccsds_asm_correlator.sv	clk_tc_safe	0, 1, 2	Soft-LLR $0^\circ/180^\circ$ correlator for CLTU, TM/AOS, Proximity-1, and LDPC ASMs.
rtl/ccsds_bch6356_dec.sv	clk_tc_safe	0, 2	Spacecraft expurgated BCH$(63, 56)$ SEC-DED decoder and CLTU tail detector.
rtl/ccsds_bch6356_enc.sv	clk_sys	1, 2	Ground EGSE BCH$(63, 56)$ CLTU encoder with 0x55 fill-octet padding.
rtl/ccsds_ul_short_ldpc_dec.sv	clk_tc_safe	0, 2	4-phase 16-row/clk CCSDS 231.1-O-1 $(128, 64)$ Soft-OMS iterative LDPC decoder.
rtl/ccsds_tc_uslp_deframer.sv	clk_tc_safe	0, 1, 2	Uplink TC (0x55 fill strip), Proximity-1, and USLP deframer + CRC checker.
rtl/ccsds_farm1_tmr.sv	clk_tc_safe	0, 2	3x TMR COP-1 FARM-1 responder with Two-Phase SDLS commit/rollback.
rtl/ccsds_hcpd_ctrl.sv	clk_tc_safe	0, 2	3x TMR Emergency Command Decoder (VCID=63) driving OBC reset & relays.
rtl/ccsds_sdls_engine.sv	clk_sys	0, 1, 2	Inline AES-128-CTR/GCM ($GF(2^{128})$ 4b/clk GHASH) and 64-SA SECDED RAM.
rtl/ccsds_pkt_reassembler.sv	clk_sys	0, 1, 2	Uplink Space Packet reassembler with automatic APID 0x7FF Idle stripping.
3. Downlink Telemetry, Framing, Ground EGSE & Outer/Inner FEC Chain (9 Files)			
rtl/ccsds_vc_arbiter_mux.sv	clk_sys	0, 1, 2	8-VC priority/weighted arbiter and Cycle-0 closed-form FHP calculator.
rtl/ccsds_tm_aos_uslp_framer.sv	clk_sys	0, 1, 2	Downlink TM (8B CUC OBT), AOS (FHEC), Proximity-1, and USLP frame builder.
rtl/ccsds_dl_rs255_223_enc.sv	clk_sys	0, 2	Downlink dual-basis RS$(255, 223)$ encoder ($I \in \{1..8\}$, virtual fill Q).
rtl/ccsds_dl_rs255_223_dec.sv	clk_sys	1, 2	Ground EGSE $t = 16$ Berlekamp-Massey, Chien, and Forney RS$(255, 223)$ decoder.
rtl/ccsds_dl_turbo_ldpc_enc.sv	clk_sys	0, 2	Downlink AR4JA quasi-cyclic LDPC and 16-state RSC Turbo ($r = 1/2..1/6$) encoder.
rtl/ccsds_conv_viterbi_codec.sv	clk_tx_sym	0, 1, 2	Rate-$1/2$, $K = 7$ convolutional encoder (Tx) and 64-state soft Viterbi decoder (Rx).
rtl/ccsds_fop1_engine.sv	clk_sys	1, 2	Ground EGSE COP-1 FOP-1 initiator with Go-Back-N retransmission queue.
rtl/ccsds_tc_cltu_framer.sv	clk_sys	1, 2	Ground EGSE TC Transfer Frame builder (40b Primary + 8b Segment + CRC-16).
rtl/ccsds_tm_aos_deframer.sv	clk_sys	1, 2	Ground EGSE TM/AOS deframer and CRC-gated 32-bit CLCW extractor.
4. Proximity-1 ARQ, Baseband RF Modem, PN Ranging & Top-Level Wrapper (4 Files)			
rtl/ccsds_prox1_plcw_ctrl.sv	clk_sys	0, 1, 2	Proximity-1 COP-P 32-bit PLCW sequence and retransmission controller.
rtl/ccsds_pn_ranging.sv	clk_adc_iq	0, 1, 2	Regenerative T2B/T4B 6-component PN ranging correlator ($L = 1{,}009{,}470$).
rtl/ccsds_phy_modem.sv	clk_adc_iq	0, 1, 2	17-tap CSD 4-SPS SRRC, Gardner STR, FLL/Costas PLL, and S/X/Ka turnaround.
rtl/ccsds_sat_ttc_top.sv	All 4 Clocks	0, 1, 2	Parameterized top-level transceiver integrating all sub-cores and CDC bridges.
5. Orbital Channel Models & Self-Checking Verification Suites (2 RTL + 11 TB Files)			
rtl/ccsds_*_channel_model.sv (2)	clk_adc_iq	Sim Model	Orbital S-curve Doppler, Rician fading, AWGN, multipath, and SSPA models.
tb/tb_stage2..12_*.sv (11 files)	Verification	Testbench	11 self-checking testbenches + 34,560-trial full-chip RF matrix & pass suite.

Table 1.1: Grouped inventory of the 30 SystemVerilog modules in rtl/ and 11 verification suites in tb/.

CCSDS Book	Standard Title	Primary RTL Module(s)	Hardware Implementation Summary
1. Space Packet, Data Link Framing & Closed-Loop ARQ Protocols			
133.0-B-2 [1]	Space Packet Protocol	ccsds_vc_arbiter_mux, pkt_reasm	48-bit header check, Cycle-0 FHP segmentation, APID 0x7FF idle insert/strip.
132.0-B-3 [2]	TM Space Data Link	ccsds_tm_aos_uslp_framer/deframer	48-bit TM header, MC/VC counters, 8B CUC OBT, atomic 32-bit CLCW, CRC-16.
232.0-B-4 [3]	TC Space Data Link	ccsds_tc_uslp_deframer, cltu_framer	40-bit TC header, 8-bit Segment Header (MAP_ID), 0x55 fill strip, Type-AD/BD/BC.
232.1-B-2 [4]	COP-1 (FARM-1 / FOP-1)	ccsds_farm1_tmr, ccsds_fop1_engine	3x TMR FARM-1 (8 VCs, Two-Phase SDLS commit) and ground FOP-1 Go-Back-N.
732.0-B-4 [5]	AOS Space Data Link	ccsds_tm_aos_uslp_framer, gf256	48-bit AOS header with 16-bit $GF(2^4)$ RS$(10, 6)$ FHEC SEC-DED header protection.
732.1-B-3 [6]	Unified Space Link (USLP)	ccsds_tc_uslp_deframer, tm_framer	Truncated (32b) and Extended (56–112b) headers, 16b SCID, 24b TFDFH, CRC-16/32.
211.0/1/2-B [7]	Proximity-1 Space Link	ccsds_prox1_plcw_ctrl, crc_univ	Version-3 40-bit header, 24-bit ASM (0xFAF320), 32-bit PLCW, CRC-32 (0x00A00805).
2. Synchronization & Channel Coding (Forward Error Correction)			
131.0-B-5 [8]	TM Sync & Channel Coding	rs255_223, conv_viterbi, turbo_ldpc	Dual-basis RS$(255, 223)$ ($I \in \{1..8\}, Q$), $K = 7$ Conv/Viterbi, 16-state Turbo, h_{TM}.
231.0-B-4 [9]	TC Sync & Channel Coding	ccsds_bch6356_dec/enc, hcpd_ctrl	CLTU (0xEB90), expurgated BCH$(63, 56)$ SEC-DED, Tail (0xC5..79), h_{TC}, 3x TMR HCPD.
231.1-O-1 [10]	Uplink Short-Block LDPC	ccsds_ul_short_ldpc_dec.sv	Quasi-cyclic $(128, 64)$ LDPC with 4-phase 16-row/clk Soft-OMS iterative decoder.
3. Link Security, Timekeeping, RF Modulation & Ranging			
301.0-B-4 [11]	CUC Time Code Formats	ccsds_axi_lite_csr.sv	48-bit CUC OBT (32b Coarse + 16b Fine, P-Field 0x2E) locked to pps_1hz_in.
355.0/1-B [12]	Space Data Link Security	ccsds_sdls_engine.sv	NIST AES-128-CTR & AES-128-GCM ($GF(2^{128})$ 128b GHASH tag) + 64-SA RAM.
401.0-B-32 [13]	RF & Modulation Systems	ccsds_phy_modem.sv	BPSK/QPSK/OQPSK/PCM-PM, 17-tap 4-SPS CSD SRRC, Gardner STR, S/X/Ka ratios.
414.1-B-3 [14]	Pseudo-Noise Ranging	ccsds_pn_ranging.sv	6-component weighted-voting T2B/T4B correlator ($L = 1{,}009{,}470$) & Q-channel overlay.

Table 1.2: Grouped CCSDS Blue and Orange Book compliance matrix for ccsds_sat_ttc_top (v2.40.3).

Parameter Name	Value	Default?	Elaborated / Pruned RTL Scope	Hardware Synthesis & Resource Impact
ROLE_MODE	0	**Yes**	**Spacecraft Flight Persona** (23 modules elaborated; 5 ground modules pruned)	Synthesizes spacecraft Rx decoders (BCH, LDPC, FARM-1, HCPD) and Tx encoders (RS, Turbo). Prunes ground FOP-1, CLTU enc, and $t = 16$ RS dec.
ROLE_MODE	1	No	**Hardware Ground EGSE Persona** (22 modules elaborated; 6 flight modules pruned)	Synthesizes ground Tx (FOP-1 Go-Back-N, TC/CLTU enc) and ground Rx ($t = 16$ Berlekamp-Massey RS dec, TM/AOS deframer with CLCW extractor).
ROLE_MODE	2	No	**Universal Crosslink & BIST Persona** (All 28 synthesizable modules elaborated)	Instantiates both spacecraft flight and ground EGSE endpoints simultaneously for Proximity-1/USLP crosslinks and autonomous on-orbit loopback BIST.
ENABLE_SDLS	1'b1	**Yes**	Elaborates ccsds_sdls_engine.sv (g_sdls_engine)	Includes full NIST FIPS-197 AES-128-CTR, NIST SP 800-38D AES-128-GCM ($GF(2^{128})$ GHASH), and 64-entry Hsiao $(39, 32)$ SECDED SA RAM.
ENABLE_SDLS	1'b0	No	Elaborates zero-latency wire bypass (g_no_sdls_engine)	Physically elides the cryptographic engine and 64-SA RAM, hardwiring sdls_auth_ok = 1'b1. **Saves 7,919 Slice LUTs and 5,571 Slice FFs.**
ENABLE_UL_LDPC	1'b1	**Yes**	Elaborates ccsds_ul_short_ldpc_dec.sv (g_ul_ldpc_dec)	Includes the 4-phase 16-row/clk CCSDS 231.1-O-1 $(128, 64)$ Soft-OMS iterative uplink LDPC decoder alongside BCH$(63, 56)$.
ENABLE_UL_LDPC	1'b0	No	Prunes ccsds_ul_short_ldpc_dec.sv (g_no_ul_ldpc_dec)	Hardwires ldpc_dec_tvalid = 1'b0 for classic BCH$(63, 56)$-only missions. **Saves 12,610 Slice LUTs and 1,132 Slice FFs.**
IQ_BUS_WIDTH	16	**Yes**	Elaborates 16-bit MSB-padded container mode	Exposes 16-bit signed rx_i/q_adc_in[15:0] (slicing [11:0] internally) and tx_i/q_dac_out[15:0] (padding {phy_tx_12b, 4'b0000} into [15:4]).
IQ_BUS_WIDTH	12	No	Elaborates direct 12-bit Q1.11 mode	Exposes direct 12-bit signed rx_i/q_adc_in[11:0] and tx_i/q_dac_out[11:0] ports matching the internal ccsds_phy_modem bit width 1:1.

Table 2.1: Complete Compile-Time Synthesis Generics and Per-Value Elaboration Behavior in rtl/ccsds_sat_ttc_top.sv.

Clock Port	Nominal	Zynq-7010 ($-1L$)	Kintex US	Domain Scope, Clock Source & CDC Isolation Strategy
clk_adc_iq	120.0 MHz	135.41 MHz	271.74 MHz	Driven by the external baseband ADC/DAC or SDR transceiver sample clock. Clocks ccsds_phy_modem.sv (4-SPS SRRC, Gardner STR, Costas loop, turnaround NCO) and ccsds_pn_ranging.sv. Crosses to clk_tc_safe via u_rx_phy_cdc_fifo.
clk_tc_safe	16.0 MHz	110.22 MHz	203.38 MHz	Driven by an independent crystal oscillator dedicated to the Uplink Telecommand chain (ccsds_asm_correlator, ccsds_bch6356_dec, ccsds_ul_short_ldpc_dec, ccsds_tc_uslp_deframer, 3x TMR FARM-1, and 3x TMR HCPD). Operates even if clk_sys halts.
clk_sys	100.0 MHz	116.40 MHz	217.34 MHz	Main OBC AXI4-Lite CSR and AXI4-Stream Space Packet clock. Clocks ccsds_axi_lite_csr, ccsds_sdls_engine, ccsds_pkt_reassembler, ccsds_vc_arbiter_mux, ccsds_tm_aos_uslp_framer, and ccsds_dl_rs255_223_enc.
clk_tx_sym	50.0 MHz	169.46 MHz	232.50 MHz	Downlink symbol/CADU clock. Driven from clk_sys via the 512-byte High-Watermark Pre-Fill asynchronous SECDED FIFO (u_tx_hwm_cdc_fifo) so the convolutional encoder and modulator never experience an underflow bubble.

Table 2.2: Clock Domains, Nominal Targets, and Measured Two-Pass True $F_{\max}$ Across Slowest-Grade Zynq-7010 (-1L) and Kintex UltraScale (XQRKU060).

- **External 1-PPS Synchronization with Autonomous Internal Fallback (pps_1hz_in):** The 48-bit CCSDS 301.0-B-4 Unsegmented Time Code (OBT) counter in ccsds_axi_lite_csr.sv increments its 32-bit coarse seconds register (0x10) on each rising edge of pps_1hz_in (synchronized into clk_sys via a 2-FF ASYNC_REG bridge). If the external GNSS or atomic clock 1-PPS signal is absent or interrupted for > 1.05 s (105,000,000 cycles of clk_sys), an internal 100 MHz hardware divider automatically takes over 1 Hz coarse-second ticking while smoothly incrementing the 16-bit fine sub-seconds register (0x14, 15.26 μs resolution).

2.4 Distributed LUTRAM Footprint & Timing Closure

To eliminate Block RAM and DSP hard-macro Single-Event Functional Interrupts (SEFIs) while keeping flip-flop count under 14,000 FFs, ccsds_sat_ttc_top applies three synthesis-level design techniques (**Tables 2.3–2.4**):

- **Synchronous Distributed LUTRAM Inference (1,200 LUTRAMs):** In rtl/ccsds_cdc_fifo_ecc.sv, the 39-bit FIFO arrays (mem_ecc_reg) use a synchronous write process (always_ff @(posedge clk_wr) with no asynchronous reset), mapping all CDC and 8-VC staging buffers into RAM64M/RAM32M SLICEM primitives rather than discrete flip-flops.
- **Continuous Combinational Hsiao $(39, 32)$ SECDED Scrubbing:** Every 32-bit word read from Distributed LUTRAM is verified and corrected on the fly by a 7-bit odd-weight Hsiao SEC-DED syndrome decoder.
- **Multiplier-Free CSD Shift-Add DSP (0 DSP48):** All 17-tap SRRC filters and $GF(2^{128})$ GHASH multipliers use binary Canonical Signed-Digit (CSD) adder trees and 4-bit/clock LUT6 XOR steps.

Synthesis Profile (ROLE=0)	Total LUTs (Logic + LUTRAM)	Slice FFs	BRAM / DSP	Zynq-7020 (53.2k LUT6)	Kintex US (XQRKU060)	Lattice MachXO5-NX (LFMXO5-100T, 80k LUT4)
Full Flight (SDLS=1, LDPC=1)	**28,861** (27.6k + 1.2k)	**13,864**	**0 / 0**	**54.25%**	**8.70%**	**~46.8%** (~ 37.5k LUT4)
BCH + GCM (SDLS=1, LDPC=0)	**16,251** (15.0k + 1.2k)	**12,732**	**0 / 0**	**30.55%**	**4.90%**	**~26.4%** (~ 21.1k LUT4)
Compact (SDLS=0, LDPC=0)	**8,332** (7.1k + 1.2k)	**7,161**	**0 / 0**	**15.66%**	**2.51%**	**~13.5%** (~ 10.8k LUT4)

Table 2.3: Measured Synthesis Resource Utilization (Logic LUTs + 1,200 Distributed RAM64M LUTRAMs + Slice Flip-Flops, 0 BRAM / 0 DSP).

3 Signal Dictionary, Sideband Bus & I/Q Alignment

To integrate ccsds_sat_ttc_top into a top-level FPGA design, the hardware engineer connects seven logical signal groups: clocks and resets, the AXI4-Lite control bus, hardware mission straps, the baseband I/Q ADC/DAC interface, the AXI4-Stream OBC packet ingress and egress buses, the optional downlink CADU tap, and the emergency HCPD and fault outputs. This chapter provides the complete top-level port and signal dictionary (59 ports), defines

Clock Domain	Nominal Target	Zynq-7010 ($-1L$) True F_{max}	Kintex US True F_{max}	Logic Levels	Limiting Physical Critical Path (Vivado 2022.2 STA)
clk_sys	100.0 MHz (10.0 ns)	**116.40** MHz (8.59 ns)	**217.34** MHz (4.60 ns)	**7**	u_rx_commit_cdc_fifo/rd_ptr_q[0] → u_csr/cnt_uncorr_err_q[0]
clk_adc_iq	120.0 MHz (8.33 ns)	**135.41** MHz (7.39 ns)	**271.74** MHz (3.68 ns)	**9**	u_phy_modem/rx_phase_q16[10] → u_phy_modem/i_derot_q[13]
clk_tx_sym	50.0 MHz (20.0 ns)	**169.46** MHz (5.90 ns)	**232.50** MHz (4.30 ns)	**9**	u_tx_hwm_cdc_fifo/wr_sync2[1] → u_tx_hwm_cdc_fifo/rd_gray[6]
clk_tc_safe	16.0 MHz (62.5 ns)	**110.22** MHz (9.07 ns)	**203.38** MHz (4.92 ns)	**11**	u_ul_deframer/cur_tc_hdr[vcid] → u_farm1/clcw_scheduled_out

Table 2.4: Two-Pass Static Timing Analysis (STA) Across Slowest-Grade Zynq-7010 (-1L) and Radiation-Tolerant Kintex UltraScale (XQRKU060).

the 64-bit ccsds_tuser_t sideband bus bit-by-bit, explains how to adapt the baseband I/Q port to any converter bit width or alignment convention, and explains why uplink FEC is controlled by hardware straps rather than software registers.

3.1 Top-Level Port Reference (59 Ports)

Table 3.1 documents all 59 top-level ports of ccsds_sat_ttc_top (rtl/ccsds_sat_ttc_top.sv), grouped by functional interface.

Table 3.1: Complete Pinout Reference for ccsds_sat_ttc_top (v2.40.3).

Port Name	Dir	Width	Clock	Functional Description & Integration Notes
1. Clocks, Reset & 1-PPS Time Synchronization Interface				
clk_sys	In	1	clk_sys	System packet, framing, CSR, SDLS & outer FEC clock (100 MHz nominal).
clk_tc_safe	In	1	clk_tc_safe	Independent radiation-safe telecommand & HCPD clock (16 MHz nominal).
clk_tx_sym	In	1	clk_tx_sym	Downlink symbol & inner coding clock (50 MHz nominal).
clk_adc_iq	In	1	clk_adc_iq	Baseband RF ADC/DAC & PN ranging sample clock (120 MHz nominal).
rst_n	In	1	Async	Active-low master reset; hold low ≥ 8 clk_tc_safe cycles (≥ 500 ns).
pps_1hz_in	In	1	Async	External 1 Hz (1-PPS) pulse for 48-bit CUC OBT; auto-fallback if idle > 1.05 s.
2. AXI4-Lite Control & Status Register (CSR) Slave Interface (0x00..0x50)				
s_axi_awvalid / s_axi_awready	In/Out	1, 1	clk_sys	AXI4-Lite write address valid/ready handshake.
s_axi_awaddr	In	8	clk_sys	AXI4-Lite byte write address (0x00..0x50, 32-bit word aligned).
s_axi_wvalid / s_axi_wready	In/Out	1, 1	clk_sys	AXI4-Lite write data valid/ready handshake.
s_axi_wdata / s_axi_wstrb	In	32, 4	clk_sys	32-bit write data bus and 4-bit byte-lane write strobe (4'hF).
s_axi_bvalid / s_axi_bready	Out/In	1, 1	clk_sys	AXI4-Lite write response valid/ready handshake.
s_axi_bresp	Out	2	clk_sys	Write response status (2'b00 = OKAY).
s_axi_arvalid / s_axi_arready	In/Out	1, 1	clk_sys	AXI4-Lite read address valid/ready handshake.
s_axi_araddr	In	8	clk_sys	AXI4-Lite byte read address (0x00..0x50).
s_axi_rvalid / s_axi_rready	Out/In	1, 1	clk_sys	AXI4-Lite read data valid/ready handshake.
s_axi_rdata / s_axi_rresp	Out	32, 2	clk_sys	32-bit read data bus and read response status (2'b00 = OKAY).
3. Hardware Mission Straps & Physical Mode Overrides				
link_proto_override_en	In	2	clk_sys	2'b00: Use CSR 0x04[14:12]; non-zero overrides protocol mode in hardware.
ul_fec_override_en	In	2	clk_tc_safe	Uplink FEC strap: 2'd0=Uncoded, 2'd1=BCH$(63, 56)$, 2'd2=LDPC$(128, 64)$.
sdls_en_cfg / bc_require_sdls_cfg	In	1, 1	clk_sys	Mission security straps OR-ed with CSR 0x04[26] and 0x04[27].
pn_ranging_en_cfg / pn_t4b_mode_cfg	In	1, 1	clk_adc_iq	Enable regenerative PN ranging (1) and select T4B (1) vs. T2B (0) voting.
turnaround_xband_cfg	In	1	clk_adc_iq	Selects S-Band 240/221 (0) or X-Band 880/749 (1) when X→Ka CSR is 0.
mod_mode_cfg / line_code_cfg	In	2, 2	clk_adc_iq	Modulation (0=BPSK, 1=QPSK, 2=OQPSK, 3=PCM/PM) & Line Code (0..2).
4. Uplink Baseband RF I/Q ADC & Direct Symbol Bypass Inputs				
rx_i_adc_in / rx_q_adc_in	In	W, W	clk_adc_iq	Signed two's-complement I/Q ADC bus (W = IQ_BUS_WIDTH, default 16).
rx_iq_valid_in	In	1	clk_adc_iq	Sample valid qualifier (supports decimated symbol rates div $= 1, 2, 4, 8$).
s_axis_ul_sym_tvalid / tready	In/Out	1, 1	clk_tc_safe	Direct baseband symbol/byte bypass ingress handshake (tie tvalid=0 for RF).
s_axis_ul_sym_tdata / tlast	In	8, 1	clk_tc_safe	Signed 4-bit LLR (tdata[3:0]) or uncoded frame byte + end-of-frame strobe.
5. OBC AXI4-Stream Space Packet Egress (m_axis_rx_pkt) & Ingress (s_axis_tx_pkt)				
m_axis_rx_pkt_tvalid / tready	Out/In	1, 1	clk_sys	Reassembled & authenticated Uplink Space Packet egress handshake to OBC.

Port Name	Dir	Width	Clock	Functional Description & Integration Notes
m_axis_rx_pkt_tdata / tlast	Out	8, 1	clk_sys	Uplink Space Packet byte stream; tlast=1 marks final octet of packet.
m_axis_rx_pkt_tuser	Out	64	clk_sys	Packed ccsds_tuser_t metadata (vcid, map_id, spi_idx, error flags; **Table 3.2**).
s_axis_tx_pkt_tvalid / tready	In/Out	1, 1	clk_sys	Downlink Telemetry Space Packet ingress handshake from OBC.
s_axis_tx_pkt_tdata / tlast	In	8, 1	clk_sys	Downlink Space Packet byte stream; tlast=1 required on final packet octet.
s_axis_tx_pkt_tuser	In	64	clk_sys	Ingress metadata (tuser[47:42] = vcid selects target Virtual Channel 0..7).
6. Downlink CADU Byte Stream & Baseband RF I/Q DAC Outputs				
m_axis_dl_cad_tvalid / tready	Out/In	1, 1	clk_tx_sym	Scrambled & outer-FEC encoded Downlink CADU byte stream (for external PHY).
m_axis_dl_cad_tdata / tlast	Out	8, 1	clk_tx_sym	Downlink CADU octet and end-of-CADU boundary strobe.
tx_i_dac_out / tx_q_dac_out	Out	W, W	clk_adc_iq	Signed two's-complement I/Q DAC outputs (W = IQ_BUS_WIDTH).
tx_iq_valid_out	Out	1	clk_adc_iq	Output sample valid qualifier to external DAC / SDR interface bridge.
7. Emergency Hardware Command Pulse Decoder (HCPD), Ranging & EDAC Outputs				
hcpd_obc_reset_pulse	Out	1	clk_tc_safe	Single-cycle pulse on VCID=63 MAGIC_OBC_HARD_RESET (0xEB90_A55A_C33C_F00F).
hcpd_safe_mode_force	Out	1	clk_tc_safe	Latched safe-mode level on VCID=63 MAGIC_SAFE_MODE_FORCE (0xEB90_5AA5_3CC3_0FF0).
hcpd_rf_chain_select	Out	2	clk_tc_safe	Redundant RF chain relay select (2'b01=Chain A, 2'b10=Chain B).
pn_ranging_locked_out	Out	1	clk_adc_iq	Asserts (1) when all 6 PN ranging correlator components ($C_1..C_6$) achieve lock.
fault_irq_out	Out	1	clk_sys	Level interrupt asserted when uncorrectable FEC/CRC or SDLS MAC counter > 0.
edac_scrub_req_out	Out	1	clk_sys	Single-cycle pulse on any 3x TMR SEU correction or SECDED scrub event.

3.2 The 64-Bit ccsds_tuser_t AXI4-Stream Sideband Bus

Both the uplink packet egress port (m_axis_rx_pkt_tuser) and downlink packet ingress port (s_axis_tx_pkt_tuser) carry the 64-bit packed structure ccsds_tuser_t defined in rtl/ccsds_pkg.sv (**Table 3.2**). On the AXI4-Stream packet interface, tuser[10:4] encodes the resolved **6-bit SA Table Index (0..63)** plus a 1-bit match-valid flag from the 64-entry Security Association RAM.[2]

Bit Slice	Struct Field	Width	Direction & Functional Meaning
[63:48]	scid	16	Spacecraft ID extracted on Rx (10b TM/TC/Prox-1, 8b AOS, 16b USLP).
[47:42]	vcid	6	**Required on Tx ingress** to select Virtual Channel (0..7); reports frame VCID on Rx egress.
[41:36]	map_id	6	Multiplexer Access Point ID (TC Segment Header / USLP) or Proximity-1 Port ID.
[35:33]	proto_mode	3	Protocol identifier (0=TM, 1=TC, 2=AOS, 3=Proximity-1, 4=USLP).
[32:29]	fec_mode	4	Active FEC codec identifier (ccsds_fec_mode_e).
[28:23]	frame_flags	6	[28] bypass_flag, [27] ctrl_cmd_flag, [26] sec_en, [25] ocf_en, [24] fecf_en, [23] fecf_crc32.
[22:19]	rs_interleave	4	Active Reed-Solomon interleaving depth $I \in \{1..8\}$.
[18:11]	rs_virt_fill_q	8	Active Reed-Solomon implicit virtual fill byte count $Q \in [0..200]$.
[10:4]	spi	7	Resolved 6-bit Security Association (SA) RAM index ([9:4] $\in [0..63]$) + match flag ([10]) from the 64-entry SECDED SA table (the raw 16-bit header SPI is available on internal submodule net u_sdls.rx_active_spi_out[15:0]).
[3:0]	err_flags	4	Rx diagnostic flags: [3] crc_err, [2] fec_uncorr_err, [1] sdls_mac_err, [0] pkt_len_err.

Table 3.2: Bit-Level Layout of the 64-Bit ccsds_tuser_t AXI4-Stream Sideband Bus (rtl/ccsds_pkg.sv).

3.3 Baseband Digital I/Q Bus Alignment

Different analog-to-digital converters, digital-to-analog converters, and vendor SDR interface IP cores pack sub-16-bit two's-complement I/Q samples into 12-bit, 14-bit, or 16-bit words using either LSB-aligned (sign-extended) or MSB-aligned (left-justified) conventions. Internally, the baseband DSP modem (rtl/ccsds_phy_modem.sv) operates on signed 12-bit Q1.11 samples (phy_rx_i_12b[11:0] and phy_tx_i_12b[11:0]).

Out of the box, rtl/ccsds_sat_ttc_top.sv (lines 119–134) provides two compile-time port configurations controlled by parameter int unsigned IQ_BUS_WIDTH:

[2]The raw 16-bit wire-level Security Parameter Index (SPI) extracted from the SDLS Security Header is provided on internal submodule port u_sdls.rx_active_spi_out[15:0] (not exposed on the top-level entity boundary) and is provisioned via CSR 0x4C.

1. **Direct 12-Bit Native Mode (`IQ_BUS_WIDTH = 12`):** Both `rx_i/q_adc_in[11:0]` and `tx_i/q_dac_out[11:0]` are 12-bit ports wired 1:1 to the internal modem:

$$\begin{aligned} \texttt{phy_rx_i_12b[11:0]} &= \texttt{rx_i_adc_in[11:0]}, \\ \texttt{tx_i_dac_out[11:0]} &= \texttt{phy_tx_i_12b[11:0]}. \end{aligned} \tag{3.1}$$

2. **16-Bit Container Mode (`IQ_BUS_WIDTH = 16`, Default):** Designed for 16-bit AXI/DMA converter interfaces where incoming Rx samples are sign-extended in `[11:0]` and outgoing Tx samples are expected MSB-aligned in `[15:4]`:

$$\begin{aligned} \texttt{phy_rx_i_12b[11:0]} &= \texttt{rx_i_adc_in[11:0]}, \\ \texttt{tx_i_dac_out[15:0]} &= \{\texttt{phy_tx_i_12b[11:0]}, \texttt{4'b0000}\}. \end{aligned} \tag{3.2}$$

Adapting to Other ADC/DAC Bit Widths or MSB-Aligned Serializers: If your target RF transceiver or ADC outputs 14-bit or 16-bit two's-complement Rx samples left-justified (MSB-aligned) in `[15:0]`—as occurs on raw 16-bit LVDS Source-Synchronous Interfaces (`SSI`)—simply arithmetic-right-shift the incoming 16-bit Rx word by 4 bits in your top-level FPGA wrapper before driving `rx_i_adc_in` (or update line 124 of `rtl/ccsds_sat_ttc_top.sv` to `assign phy_rx_i_12b = rx_i_adc_in[15:4];`). Because `tx_i_dac_out[15:0]` already places the 12-bit Tx waveform in `[15:4]`, the transmit bus requires no adjustment.

3.4 Hardware Mission Straps vs. Software CSRs

When reviewing the top-level port list in **Table 3.1**, integrators often notice that **Downlink FEC is software-programmable via CSR `0x04[19:16]`**, whereas **Uplink FEC is controlled by dedicated hardware input pins (`ul_fec_override_en[1:0]`)**. This asymmetry is an intentional spacecraft survival feature:

Spacecraft "Chicken-and-Egg" Deadlock Prevention: Downlink telemetry originates from the On-Board Computer (OBC); flight software must already be executing to generate telemetry packets, making software CSR control safe and flexible. Conversely, the uplink telecommand chain is the spacecraft's emergency recovery lifeline. If the OBC hangs, crashes into a boot-ROM fault, or suffers an uncorrectable memory upset, flight software cannot execute AXI4-Lite register writes. Driving `ul_fec_override_en[1:0]` from non-volatile board straps (or radiation-hardened supervisor GPIOs) in the independent 16 MHz `clk_tc_safe` domain guarantees that uplink command decoding is active within nanoseconds of power-on reset.

4 Functional Subsystem Operation & Protocol Configuration

With the external signals and clock domains established, this chapter summarizes how the internal protocol engines operate across the five Space Data Link framing formats (**Table 4.1**), the closed-loop `COP-1` (`CLCW`) and `Proximity-1 COP-P` (`PLCW`) ARQ control words (**Table 4.2**), and the Two-Phase `FARM-1/SDLS AES-128-GCM` security architecture (**Figure 4.1**). Complete mathematical derivations of these blocks are provided in Volume II, Chapters 2–5.

Protocol Mode	TFVN	Prim. Header	SCID / VCID	FECF CRC	Key Hardware Features in `ccsds_sat_ttc_top`
TM (`132.0-B-3`)	`2'b00`	48 bits (6 B)	10b / 3b (8 VCs)	CRC-16 (`0x1021`)	Cycle-0 `FHP`, optional 8B CUC `OBT` Sec. Header, 32-bit `CLCW` OCF.
TC (`232.0-B-4`)	`2'b00`	40 bits (5 B)	10b / 6b (64 VCs)	CRC-16 (`0x1021`)	Optional 8b Segment Header (`MAP_ID`), `0x55` CLTU fill strip, COP-1 `FARM-1`.
AOS (`732.0-B-4`)	`2'b01`	48b + 16b FHEC	8b / 6b (64 VCs)	CRC-16 (`0x1021`)	Inline $GF(2^4)$ RS$(10, 6)$ `FHEC` SEC-DED header protection, 24b VC counter.
Prox-1 (`211.0-B`)	`2'b10`	40 bits (5 B)	10b / 3b Port	CRC-32 (`0x00A00805`)	24-bit ASM (`0xFAF320`), bidirectional COP-P ARQ via 32-bit `PLCW`.
USLP (`732.1-B-3`)	`4'b1100`	32b Trunc / 56–112b Ext	16b / 6b + 4b MAP	CRC-16 or CRC-32	Supports both low-overhead 4B Truncated headers and Extended headers + 24b `TFDFH`.

Table 4.1: Comparison of the Five Synthesizable Space Data Link Framing Modes.

4.1 Closed-Loop COP-1/COP-P ARQ & Speculative SDLS

In a conventional receiver where FARM-1 updates its sequence counter $V(R)$ before cryptographic verification, an adversary transmitting unauthenticated frames with valid CRCs and incremented $N(S)$ values can push FARM-1 into Lockout. To prevent this Denial-of-Service vector, ccsds_sat_ttc_top couples ccsds_farm1_tmr.sv and ccsds_sdls_engine.sv through a **Two-Phase Speculative Commit/Rollback Pipeline** (**Figure 4.1B**):

1. **Phase A — Speculative Window Pre-Check (clk_tc_safe):** ccsds_farm1_tmr.sv checks whether the incoming frame sequence number $N(S)$ falls within the sliding window while keeping $V(R)$ frozen, and streams the frame into the speculative rollback FIFO (u_rx_commit_cdc_fifo).
2. **Phase B — Cryptographic Tag Verification (clk_sys):** ccsds_sdls_engine.sv looks up the 16-bit SPI in its 64-entry SECDED SA RAM, checks the 32-bit anti-replay sequence number, and verifies the 128-bit AES-128-GCM GHASH tag.
3. **Atomic Commit or Rollback:** If the tag is valid, sdls_commit_pulse advances $V(R)$, updates the 32-bit CLCW (**Table 4.2**), and releases the frame to ccsds_pkt_reassembler.sv; if verification fails, sdls_rollback_pulse rewinds the FIFO write pointer and leaves $V(R)$ untouched.

Bit Slice	COP-1 CLCW (0x20)	Proximity-1 PLCW (0x24)	Functional Meaning & Closed-Loop Ground Action
[31]	ctrl_word_type = 0	ctrl_word_type = 0	Identifies valid 32-bit Operational Control Field report word.
[30:29]	clcw_version = 00	plcw_version = 00	CCSDS 232.1-B-2 / 211.0-B-6 Version-1 control word format.
[28:26]	status_field[2:0]	status_field[2:0]	Mission status bits (b0 = Uplink Short-LDPC active indicator).
[25:24]	cop_in_effect = 01	cop_in_effect = 10	01 = COP-1 active; 10 = Proximity-1 COP-P active.
[23:18]	vcid[5:0]	pcid / vcid[5:0]	Virtual Channel ID to which this sequence report applies.
[17:16]	rsvd_spare = 00	rsvd_spare = 00	Reserved spare bits (always driven to 2'b00).
[15:11]	no_rf, no_bit, lock, wait, retx	no_rf, no_bit, lock, wait, retx	[13] Lockout (requires Type-BC Unlock); [11] Retransmit (triggers Go-Back-N).
[10:8]	farm_b_counter[1:0], 0	expedited_cnt[2:0]	Increments on each accepted Type-BD / Type-BC bypass command frame.
[7:0]	report_value_vr[7:0]	report_value_vr[7:0]	Next expected in-order sequence number $V(R) \in [0..255]$.

Table 4.2: Bitfield layout of the 32-bit COP-1 CLCW (CCSDS 232.1-B-2) and Proximity-1 PLCW (CCSDS 211.0-B-6).

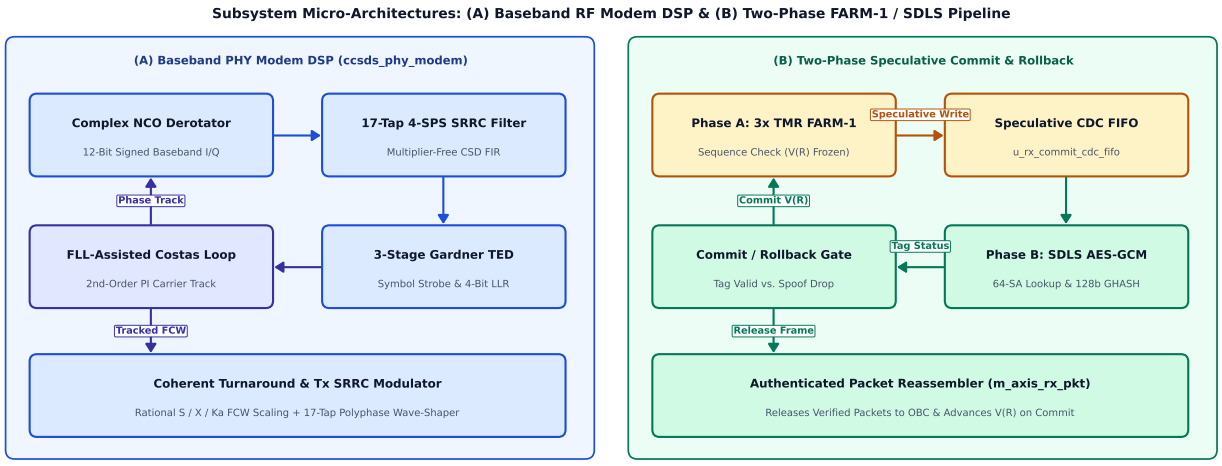


Figure 4.1: Baseband RF modem DSP pipeline (ccsds_phy_modem.sv) and Two-Phase FARM-1/SDLS commit/rollback architecture.

5 Programmer's Register Map (0x00–0x50)

All software interaction with ccsds_sat_ttc_top takes place over the 32-bit AXI4-Lite slave interface (rtl/ccsds_axi_lite_csr.sv). This chapter provides the complete address map from offset 0x00 through 0x50 (**Table 5.1**) followed by bit-by-bit field definitions for the mission configuration, timekeeping, fault telemetry, and SDLS cryptographic key/SA provisioning registers (**Tables 5.2–5.3**).

5.1 Register Address Space Overview

To protect mission configuration against radiation Single-Event Upsets (SEUs), all writable control registers (0x04, 0x08, 0x0C) are implemented as **3x Triple-Modular Redundant (TMR)** flip-flop banks (*_a_q, *_b_q, *_c_q) with continuous combinational majority voting and autonomous single-cycle SEU scrubbing.

Offset	Register Name	Access	Reset Value	Functional Summary
0x00	CORE_ID_VER	RO	0xCC5D_0230	[31:16] Magic ID (0xCC5D), [15:0] Release Version (0x0230).
0x04	CTRL_SCID_PROTO	RW (3x TMR)	0x0130_0155	Master protocol, downlink FEC, OCF/FECF, SDLS, SRRC, Gardner, and X→Ka config (**Table 5.2**).
0x08	CTRL_USLP_RS_CFG	RW (3x TMR)	0x0001_4255	16-bit USLP SCID, RS$(255, 223)$ (I, Q), AES-128-GCM mode, FLL assist, TM randomizer (**Table 5.3**).
0x0C	CTRL_COP1_TFDF_CFG	RW (3x TMR)	0x010D_F010	COP-1 sliding window half-width W, TFDF byte length, and Fault IRQ enable (**Table 5.3**).
0x10	OBT_COARSE_SEC	RW	0x6530_0000	CCSDS 301.0-B-4 CUC On-Board Time 32-bit Coarse Seconds counter (obt_cuc_48b[47:16]).
0x14	OBT_FINE_SUBSEC	RW	0x0000_0000	[15:0] CUC On-Board Time 16-bit Fine Sub-Seconds counter (2^{-16} s $= 15.26$ μs LSB).
0x18	EDAC_CORR_CNTS	RO / W1C	0x0000_0000	[15:0] SECDED/BCH 1-bit corrected count (W1C bit 0); [31:16] 3x TMR SEU scrub count (W1C bit 16).
0x1C	FAULT_ERR_CNTS	RO / W1C	0x0000_0000	[15:0] Uncorrectable FEC/CRC/Pkt error count (W1C bit 0); [31:16] SDLS MAC error count (W1C bit 16).
0x20	LATEST_CLCW_MON	RO	0x0000_0000	Live 32-bit COP-1 CLCW monitor ([7:0] = $V(R)$, [11] = Retransmit, [13] = Lockout).
0x24	LATEST_PLCW_MON	RO	0x0000_0000	Live 32-bit Proximity-1 COP-P PLCW monitor ([7:0] = $V(R)$ report value).
0x28..0x44	REG_SDLS_KEY_0..7	RW	0x1C1D..0203	Eight 32-bit words forming the 256-bit SDLS Master Key (0x28=LSW [31:0] .. 0x44=MSW [255:224]).
0x48	REG_SDLS_SA_CTRL	RW	0x0001_0001	[15:0] Default Tx SPI (0x0001); [21:16] SA RAM write address (6'd1); [31] Self-clearing SA write strobe.
0x4C	REG_SDLS_SA_SPI	RW	0x0000_0001	[15:0] 16-bit Security Parameter Index (SPI) to write into SA RAM entry SA_WR_ADDR.
0x50	REG_SDLS_SA_SEQ	RW	0x0000_0000	[31:0] Initial 32-bit Anti-Replay Sequence Number to write into SA RAM entry SA_WR_ADDR.

Table 5.1: Complete AXI4-Lite Register Map (0x00..0x50) in rtl/ccsds_axi_lite_csr.sv.

5.2 Control & Security Register Bitfields

Table 5.2 and **Table 5.3** specify the individual bitfields of the primary control registers (0x04, 0x08, 0x0C) and the Security Association provisioning control register (0x48). Note how the hardware reset constant RESET_SCID_PROTO = 32'h0131_5155 in rtl/ccsds_axi_lite_csr.sv decodes bit-for-bit in **Table 5.2**; at boot, flight software invokes ccsds_ttc_hal_configure() to program validated mission protocol and FEC settings.

6 Bare-Metal ISO C11 HAL SDK & Software Recipes

To eliminate error-prone manual bitmask manipulation in flight software, the repository includes a zero-dependency ISO C11 Hardware Abstraction Layer (sw/ccsds_ttc_hal.h and sw/ccsds_ttc_hal.c). Designed for safety-critical flight operating systems (RTEMS, FreeRTOS, NASA cFS, Zephyr) and bare-metal bootloaders on RISC-V, ARM Cortex-R, and LEON3/4 processors, the SDK requires only standard freestanding C headers (<stdint.h>, <stdbool.h>, <stddef.h>) and uses zero dynamic memory allocation. This chapter provides a detailed function-by-function API reference, explains the eight pre-commit CCSDS Blue Book configuration gating rules, and presents complete production flight software recipes.

Bits	Field Name	Reset	Register 0x04: CTRL_SCID_PROTO (RW, 3x TMR, Reset 0x0130_0155) Description
[31:30]	TURNAROUND_KABAND	2'b00	00: Use turnaround_xband_cfg strap (S-Band 240/221 or X-Band 880/749); 01: X→Ka Cross-Band Deep-Space Cat-B (3344/749); 10/11: X→Ka Cross-Band Near-Earth Cat-A (3360/749 / 3328/749).
[29]	GARDNER_TED_EN	1'b0	1: Enables closed-loop 3-stage 4-SPS Gardner timing error detector (SRRC_4SPS_EN must be 1).
[28]	SRRC_4SPS_EN	1'b0	1: Enables 17-tap CSD 4-SPS Square-Root Raised-Cosine ($\alpha = 0.35$) Tx/Rx matched filters.
[27]	BC_REQ_SDLS	1'b0	0: Exempts COP-1 Type-BC (Unlock/Set V(R)) frames from SDLS per CCSDS 355.0-B-2 §3.3.2.3; 1: Enforces SDLS on all frames.
[26]	SDLS_EN	1'b0	1: Enables inline ccsds_sdls_engine encryption/authentication (OR-ed with sdls_en_cfg pin).
[25]	OBT_SEC_HDR_EN	1'b0	1: Inserts 8-byte CCSDS 301.0-B-4 CUC On-Board Time Secondary Header into downlink TM frames.
[24]	TC_SEG_HDR_EN	1'b1	1: Expects/generates the 8-bit TC Segment Header (Seq_Flags[1:0], MAP_ID[5:0]) in PROTO_TC.
[23]	AOS_FHEC_EN	1'b0	1: Enables 16-bit $GF(2^4)$ RS$(10, 6)$ Frame Header Error Control in PROTO_AOS / PROTO_USLP.
[22]	FECF_CRC32	1'b0	0: 16-bit CRC-16-CCITT (0x1021); 1: 32-bit Proximity-1/USLP CRC-32 (0x00A00805).
[21]	FECF_EN	1'b1	1: Appends/verifies Frame Error Control Field (CRC-16/32) on all Transfer Frames.
[20]	OCF_EN	1'b1	1: Appends 32-bit Operational Control Field (CLCW or PLCW) to downlink frames.
[19:16]	FEC_MODE	4'h0	Downlink FEC select (0=Uncoded [default at reset], 1=Reserved_Uncoded, 4=Conv $K = 7$, 5=RS$(255, 223)$, 7=Concat RS+Conv, 8=AR4JA, 10..13=Turbo).
[14:12]	PROTO_MODE	3'd0	Active link protocol (0=TM default at reset, 1=TC, 2=AOS, 3=Proximity-1, 4=USLP).
[15], [11:10]	RESERVED	3'b000	Reserved bits, read as 0.
[9:0]	SCID_10B	10'h155	10-bit Spacecraft Identifier for TM, TC, and Proximity-1 (lower 8 bits used for AOS).

Table 5.2: Exact Bit-by-Bit Specification for Register 0x04 (CTRL_SCID_PROTO, Reset 0x0130_0155).

Reg	Bits	Field Name	Reset	Registers 0x08, 0x0C, and 0x48..0x50 Bitfield Descriptions
0x08	[30]	TM_RAND_EN_CFG	1'b0	1: Enables downlink CCSDS 131.0-B-5 $h_{\text{TM}}(x) = x^8 + x^7 + x^5 + x^3 + 1$ CADU pseudo-randomizer.
0x08	[29]	FLL_ASSIST_EN	1'b0	1: Enables sign-quantized Differential Cross-Product FLL discriminator for wide-CFO pull-in (± 640 steps).
0x08	[28]	SDLS_GCM_MODE	1'b0	0: AES-128-CTR mode; 1: Full NIST SP 800-38D AES-128-GCM with 128-bit $GF(2^{128})$ GHASH tag.
0x08	[27:20]	RS_VIRT_FILL_Q	8'd0	Implicit virtual fill length $Q \in [0..200]$ bytes per RS$(255, 223)$ codeword (shortens message to $223 - Q$).
0x08	[19:16]	RS_INTERLEAVE_I	4'd1	Reed-Solomon interleaving depth $I \in \{1, 2, 3, 4, 5, 8\}$ codewords per CADU.
0x08	[15:0]	USLP_SCID_16B	16'h4255	16-bit Spacecraft Identifier inserted into CCSDS 732.1-B-3 USLP Primary Headers.
0x0C	[24]	FAULT_IRQ_EN	1'b1	1: Enables fault_irq_out assertion when FAULT_ERR_CNTS (0x1C) is non-zero.
0x0C	[18:8]	TFDF_BYTE_LEN	11'd223	Transfer Frame Data Field byte length ($\in [1..1024]$, set to $223 - Q$ − Header/Trailer Overhead for RS).
0x0C	[7:0]	WINDOW_HALF_W	8'd16	COP-1 FARM-1 positive/negative sliding window half-width $W \in [1..128]$.
0x48	[31]	SA_WR_EN	1'b0	Self-clearing write strobe: writing 1 commits REG_SDLS_SA_SPI (0x4C) and SEQ (0x50) into SA RAM [21:16].
0x48	[21:16]	SA_WR_ADDR	6'd1	Target 6-bit VCID / SA table index (0..63) in the Hsiao $(39, 32)$ SECDED Security Association RAM.
0x48	[15:0]	TX_DEFAULT_SPI	16'd1	Default 16-bit Security Parameter Index (SPI) stamped into outgoing SDLS Security Headers.

Table 5.3: Bitfield Specifications for CTRL_USLP_RS_CFG (0x08), CTRL_COP1_TFDF_CFG (0x0C), and REG_SDLS_SA_CTRL (0x48).

6.1 HAL API Function Reference (sw/ccsds_ttc_hal.h)

Each of the twelve public functions in sw/ccsds_ttc_hal.h is documented below in detail with its exact C prototype, parameter semantics, hardware register transactions, and return status codes.

```
int ccsds_ttc_hal_init(ccsds_ttc_dev_t *dev, volatile uint32_t *base_addr);
```

Purpose: Initializes a driver instance handle and verifies hardware presence on the AXI4-Lite bus before allowing any configuration writes.

Parameters: dev *[out]*: Pointer to caller-allocated ccsds_ttc_dev_t device context structure; base_addr *[in]*: Physical or virtual MMIO base address of ccsds_sat_ttc_top.

Hardware Registers & Side Effects: Performs a 32-bit volatile read from offset 0x00 (CORE_ID_VER) and checks that the returned word equals 0xCC5D0230UL (CCSDS_EXPECTED_VERSION_ID).

Return Value: CCSDS_HAL_OK (0) on success; CCSDS_HAL_ERR_NULL_PTR (-1) if dev or base_addr is NULL; CCSDS_HAL_ERR_BAD_VERSION (-2) if the hardware version register does not match 0xCC5D0230UL.

```
int ccsds_ttc_hal_validate_config(const ccsds_ttc_config_t *cfg);
```

Purpose: Audits a candidate transceiver configuration structure against all eight cross-layer CCSDS Blue Book compatibility rules in software without touching hardware registers.

Parameters: cfg *[in]*: Pointer to a populated ccsds_ttc_config_t configuration structure.

Hardware Registers & Side Effects: Pure software validation function; performs zero MMIO reads or writes. Used both standalone and as the pre-commit gate inside ccsds_ttc_hal_configure().

Return Value: CCSDS_HAL_OK (0) if valid, or a specific negative error code (-1 or -10..-17; see Subsection 6.2) identifying the exact CCSDS rule violation.

```
int ccsds_ttc_hal_configure(const ccsds_ttc_dev_t *dev, const ccsds_ttc_config_t
*cfg);
```

Purpose: Validates and atomically commits a complete protocol, FEC, PHY, COP-1, and SDLS configuration to the 3x TMR hardware control registers.

Parameters: dev *[in]*: Initialized device handle; cfg *[in]*: Desired transceiver configuration structure.

Hardware Registers & Side Effects: First calls ccsds_ttc_hal_validate_config(cfg). If and only if validation returns 0, packs the bitfields and writes 0x04 (CTRL_SCID_PROTO), 0x08 (CTRL_USLP_RS_CFG), and 0x0C (CTRL_COP1_TFDF_CFG). If validation fails, hardware registers remain untouched.

Return Value: CCSDS_HAL_OK (0) on commit, or the negative validation error code (-1, -10..-17).

```
void ccsds_ttc_hal_get_config(const ccsds_ttc_dev_t *dev, ccsds_ttc_config_t *cfg_
out);
```

Purpose: Reads back the currently active 3x TMR voted configuration registers from hardware and unpacks them into a C structure for telemetry verification or state auditing.

Parameters: dev *[in]*: Initialized device handle; cfg_out *[out]*: Destination structure populated with active hardware settings.

Hardware Registers & Side Effects: Executes three 32-bit reads from offsets 0x04, 0x08, and 0x0C (which return the combinational majority-voted output of the triplicated TMR banks).

Return Value: void (no-op if either pointer is NULL).

```
int ccsds_ttc_hal_set_sdls_key256(const ccsds_ttc_dev_t *dev, const uint32_t key_
words[8]);
```

Purpose: Loads a 128-bit or 256-bit cryptographic Master Key into the SDLS engine for AES-128-CTR and AES-128-GCM authenticated encryption and decryption.

Parameters: dev *[in]*: Initialized device handle; key_words[8] *[in]*: Array of eight 32-bit words (key_words[0] = LSW [31:0] through key_words[7] = MSW [255:224]).

Hardware Registers & Side Effects: Writes offsets 0x28, 0x2C, 0x30, 0x34, 0x38, 0x3C, 0x40, and 0x44 (REG_

SDLS_KEY_0..7).

Return Value: CCSDS_HAL_OK (0) on success; CCSDS_HAL_ERR_NULL_PTR (-1) if either pointer is NULL.

```
int ccsds_ttc_hal_provision_sdls_sa(const ccsds_ttc_dev_t *dev, uint8_t vcid_addr,
uint16_t spi, uint32_t seq_low, uint16_t tx_default_spi);
```

Purpose: Provisions an entry in the 64-entry Hsiao $(39, 32)$ SECDED-protected Security Association (SA) RAM for a specific Virtual Channel (vcid_addr) and sets the default transmit SPI. To prevent GCM IV reuse across Virtual Channels or spacecraft, set spi = (scid_10b « 6) | vcid_addr.

Parameters: dev *[in]*: Device handle; vcid_addr *[in]*: SA table index (0..63); spi *[in]*: Expected 16-bit Security Parameter Index; seq_low *[in]*: Initial 32-bit anti-replay sequence number threshold; tx_default_spi *[in]*: 16-bit SPI stamped on outgoing downlink frames.

Hardware Registers & Side Effects: Writes spi to 0x4C (REG_SDLS_SA_SPI), writes seq_low to 0x50 (REG_SDLS_SA_SEQ), and writes (1U « 31) | ((vcid_addr & 0x3FU) « 16) | tx_default_spi to 0x48 (REG_SDLS_SA_CTRL), triggering a single-cycle Hsiao SECDED-encoded write into SA RAM.

Return Value: CCSDS_HAL_OK (0) on success; -1 if dev == NULL or vcid_addr > 63.

```
void ccsds_ttc_hal_get_clcw(const ccsds_ttc_dev_t *dev, ccsds_clcw_status_t *clcw_
out);
```

Purpose: Reads the latest 32-bit COP-1 Communications Link Control Word (CLCW) and unpacks all status flags and the receiver sequence counter $V(R)$.

Parameters: dev *[in]*: Device handle; clcw_out *[out]*: Pointer to ccsds_clcw_status_t receiving vcid, no_rf_avail, no_bit_lock, lockout, wait_flag, retransmit, farm_b_counter, and report_value ($V(R)$).

Hardware Registers & Side Effects: Reads offset 0x20 (LATEST_CLCW_MON). On Spacecraft (ROLE=0), reflects local FARM-1 state; on Ground EGSE (ROLE=1), reflects the latest CLCW extracted from downlink telemetry.

Return Value: void.

```
uint64_t ccsds_ttc_hal_get_obt_cuc48(const ccsds_ttc_dev_t *dev);
void ccsds_ttc_hal_set_obt_cuc48(const ccsds_ttc_dev_t *dev, uint32_t coarse_sec,
uint16_t fine_subsec);
```

Purpose: Reads or initializes the 48-bit CCSDS 301.0-B-4 Unsegmented Time Code (OBT) counter (32-bit coarse seconds in bits [47:16] and 16-bit fine sub-seconds in bits [15:0]).

Parameters: dev *[in]*: Device handle; coarse_sec *[in]*: 32-bit seconds epoch; fine_subsec *[in]*: 16-bit fractional seconds (2^{-16} s LSB).

Hardware Registers & Side Effects: Accesses offsets 0x10 (OBT_COARSE_SEC) and 0x14 (OBT_FINE_SUBSEC).

Return Value: get_obt_cuc48 returns the packed 48-bit timestamp in a uint64_t; set_obt_cuc48 returns void.

```
void ccsds_ttc_hal_get_error_telemetry(const ccsds_ttc_dev_t *dev, ccsds_err_
telemetry_t *err_out);
void ccsds_ttc_hal_clear_error_telemetry(const ccsds_ttc_dev_t *dev, bool clr_edac,
bool clr_tmr, bool clr_uncorr, bool clr_sdls);
```

Purpose: Reads or selectively clears the four saturating 16-bit hardware EDAC, TMR SEU, uncorrectable FEC/CRC, and SDLS MAC authentication error counters.

Parameters: dev *[in]*: Device handle; err_out *[out]*: Structure receiving all 4 counters; clr_* *[in]*: Booleans selecting which counters to clear via Write-1-to-Clear (W1C).

Hardware Registers & Side Effects: Reads or writes W1C strobes (bit 0 and bit 16) to offsets 0x18 (EDAC_CORR_CNTS) and 0x1C (FAULT_ERR_CNTS).

Return Value: void.

```
uint32_t ccsds_ttc_hal_reg_read(const ccsds_ttc_dev_t *dev, uint32_t reg_offset);
void ccsds_ttc_hal_reg_write(const ccsds_ttc_dev_t *dev, uint32_t reg_offset,
uint32_t val);
```

Purpose: Low-level volatile 32-bit MMIO register read and write primitives for custom diagnostic or register-test routines.

Parameters: `dev` *[in]*: Device handle; `reg_offset` *[in]*: Byte offset (`0x00..0x50`); `val` *[in]*: 32-bit word to write.

Hardware Registers & Side Effects: Directly dereferences `dev->base_addr + (reg_offset » 2)`.

Return Value: `reg_read` returns the 32-bit register value; `reg_write` returns `void`.

6.2 Pre-Commit Configuration Gating Rules

To prevent flight software from accidentally programming an illegal combination of modulation, line coding, framing protocol, and FEC codec, `ccsds_ttc_hal_validate_config()` enforces eight cross-layer compatibility rules prior to any CSR write:

1. **Modulation vs. Manchester Line Coding (`CCSDS_HAL_ERR_MOD_LINE_INCOMPAT = -10`):** Per CCSDS 401.0-B-32 [13], split-phase Manchester (`Bi-`ϕ`-L`) coding is only permitted with single-dimension carrier modulations (`BPSK` and residual-carrier `PCM/PSK/PM`). Pairing `QPSK` or `OQPSK` with `Bi-`ϕ`-L` is rejected (leaving 10 valid modulation $\times$ line-code waveform pairs).
2. **4-SPS SRRC Pulse Shaping vs. Manchester Coding (`CCSDS_HAL_ERR_SRRC_BIPHI_INCOMPAT = -11`):** Square-Root Raised-Cosine (`srrc_4sps_en = 1`) matched filtering is defined strictly for baseband `NRZ-L` and `NRZ-M` pulses; enabling SRRC alongside `Bi-`ϕ`-L` distorts the mid-bit Manchester transition and is rejected.
3. **4-SPS Gardner Timing Recovery Dependency (`CCSDS_HAL_ERR_GARDNER_REQUIRES_4SPS = -12`):** The Gardner Timing Error Detector (`gardner_ted_en = 1`) requires 4 samples per symbol from the SRRC matched filter (`srrc_4sps_en = 1`); enabling Gardner TED with `srrc_4sps_en = 0` is rejected.
4. **Protocol vs. Directional FEC Compatibility (`CCSDS_HAL_ERR_PROTO_FEC_INCOMPAT = -13`):** Rejects pairing Uplink-only Telecommand (`PROTO_TC`) with Downlink-only codecs (RS$(255, 223)$, Turbo, AR4JA), rejects pairing Downlink-only `PROTO_TM/AOS` with Uplink-only codecs (BCH$(63, 56)$, Short-LDPC), and restricts `PROTO_PROX1` to uncoded or $K = 7$ convolutional coding per CCSDS 211.2-B-3.
5. **CRC-16 vs. CRC-32 FECF Mandate (`CCSDS_HAL_ERR_CRC32_PROTO_INCOMPAT = -14`):** Enforces 16-bit CRC-16 (`fecf_crc32 = 0`) on `TM`, `AOS`, and `TC`, mandates 32-bit CRC-32 (`fecf_crc32 = 1`) on `Proximity-1`, and allows either CRC-16 or CRC-32 on `USLP`.
6. **AOS Frame Header Error Control Scope (`CCSDS_HAL_ERR_AOS_FHEC_INCOMPAT = -15`):** Rejects enabling the 16-bit $GF(2^4)$ RS$(10, 6)$ `FHEC` (`aos_fhec_en = 1`) unless the active protocol is `PROTO_AOS` or `PROTO_USLP`.
7. **Reed-Solomon Parameter Bounds (`CCSDS_HAL_ERR_RS_PARAM_RANGE = -16`):** Enforces CCSDS 131.0-B-5 interleaving depth $I \in \{1, 2, 3, 4, 5, 8\}$ and virtual fill length $Q \leq 215$.
8. **Frame Length & COP-1 Window Bounds (`CCSDS_HAL_ERR_FRAME_LEN_RANGE = -17`):** Enforces TFDF_BYTE_LEN $\in$ [1..1024], verifies that total Transfer Frame size (TFDF_BYTE_LEN + Header/OCF/FECF/SDLS Overhead) does not exceed $223 - Q$ bytes when RS$(255, 223)$ ($I = 1$) is active, and enforces COP-1 sliding window half-width $W \in$ [1..128].

6.3 Flight Software Integration Recipes

When configuring RS$(255, 223)$ ($I = 1, Q = 0$), each codeword carries exactly 223 bytes of total Transfer Frame data. With TM Primary Header (6 B) + CUC OBT Secondary Header (8 B) + SDLS Security Header (6 B) + SDLS AES-128-GCM Tag (16 B) + CLCW OCF (4 B) + CRC-16 FECF (2 B) = **42** bytes of framing and security overhead, `.tfdf_byte_len` is configured to $223 - 42 =$ **181** bytes (`181U`) so each Transfer Frame maps 1:1 into a 255-byte Reed-Solomon CADU:

```c
#include "ccsds_ttc_hal.h"

/* Recipe 1: Spacecraft Flight Boot, Domain-Separated SDLS Key/SA Load & 223B RS Frame Config */
int spacecraft_ttc_boot_init(ccsds_ttc_dev_t *dev, volatile uint32_t *mmio_base) {
```

```
    int status = ccsds_ttc_hal_init(dev, mmio_base);
    if (status != CCSDS_HAL_OK) return status;

    /* 1. Load 128-bit/256-bit Master Key into 0x28..0x44 and provision VCID=0..7 with domain-separated
     * SPI = (SCID_10B << 6) | VCID so no two Virtual Channels or spacecraft ever reuse a GCM IV */
    const uint32_t mission_key[8] = {
        0x3C4FCF09UL, 0x8815F7ABUL, 0xA6D2AE28UL, 0x16157E2BUL,
        0x00000000UL, 0x00000000UL, 0x00000000UL, 0x00000000UL
    };
    const uint16_t scid_10b = 0x0155U;
    const uint16_t default_spi = (uint16_t)((scid_10b << 6) | 0U); /* 0x5540 for SCID=0x155, VCID=0 */
    ccsds_ttc_hal_set_sdls_key256(dev, mission_key);
    for (uint8_t vcid = 0U; vcid < 8U; vcid++) {
        uint16_t vc_spi = (uint16_t)((scid_10b << 6) | vcid);
        ccsds_ttc_hal_provision_sdls_sa(dev, vcid, vc_spi, 0x00000000UL, default_spi);
    }

    /* 2. Configure Downlink TM + Concatenated RS(255,223)/Viterbi + 4-SPS SRRC/Gardner + AES-GCM.
     * Total RS(255,223) info capacity = 223 B. Overhead = 6B(TM) + 8B(OBT) + 6B(SDLS Hdr)
     * + 16B(GCM Tag) + 4B(OCF) + 2B(CRC16) = 42 B -> usable tfdf_byte_len = 223 - 42 = 181 B. */
    ccsds_ttc_config_t cfg = {
        .scid_10b = scid_10b,
        .uslp_scid_16b = 0x4255U,
        .proto_mode = CCSDS_PROTO_TM,
        .fec_mode = CCSDS_FEC_CONCAT_RS_CNV,
        .mod_mode = CCSDS_MOD_QPSK,
        .line_code = CCSDS_LINE_NRZ_L,
        .ocf_en = true, /* +4 B: Embed live 32-bit COP-1 CLCW in Downlink TM OCF */
        .fecf_en = true, /* +2 B: Append 16-bit CRC-16-CCITT FECF */
        .fecf_crc32 = false,
        .aos_fhec_en = false,
        .tc_seg_hdr_present = true,
        .obt_sec_hdr_en = true, /* +8 B: Stamp 48-bit CUC On-Board Time Secondary Header */
        .sdls_en = true, /* +6 B Header + 16 B GCM Tag: Inline CCSDS 355.0-B-2 SDLS */
        .bc_require_sdls = false, /* Allow Type-BC Unlock recovery per 355.0-B-2 3.3.2.3 */
        .window_half_w = 16U,
        .rs_interleave_i = 1U,
        .rs_virt_fill_q = 0U,
        .tfdf_byte_len = 181U, /* 181 B TFDF + 42 B Overhead = 223 B exact RS(255,223) block */
        .turnaround_kaband = CCSDS_KABAND_NONE_OR_SX, /* S-Band 240/221 via strap */
        .srrc_4sps_en = true,
        .gardner_ted_en = true,
        .sdls_gcm_mode = true, /* Full 128-bit GHASH authenticated encryption */
        .fll_assist_en = true,
        .tm_rand_en = true,
        .fault_irq_en = true
    };
    return ccsds_ttc_hal_configure(dev, &cfg);
}

/* Recipe 2: Periodic 1 Hz OBC Health ISR --- Poll CLCW, OBT & Scrub EDAC/TMR Counters */
void spacecraft_ttc_1hz_health_isr(const ccsds_ttc_dev_t *dev, ccsds_clcw_status_t *clcw,
                        ccsds_err_telemetry_t *err, uint64_t *obt_cuc48) {
    *obt_cuc48 = ccsds_ttc_hal_get_obt_cuc48(dev);
    ccsds_ttc_hal_get_clcw(dev, clcw);
    ccsds_ttc_hal_get_error_telemetry(dev, err);

    if (err->secded_bch_single_corr_cnt || err->tmr_seu_corrected_cnt ||
        err->uncorr_frame_pkt_err_cnt || err->sdls_mac_err_cnt) {
        ccsds_ttc_hal_clear_error_telemetry(dev, true, true, true, true);
    }
}
```

7 S-Band Board Integration, Wiring & Bring-Up

While **Volume II, Chapter 7** derives the S-Band RF cascade, Friis noise-temperature equations, 600 km LEO link budgets, and space-qualification rules from first principles, this chapter provides the practical **board-level hardware partitioning, pin wiring, and component bill of materials** for integrating the FPGA with an external SDR transceiver and S-Band RF Front-End (RFFE).

7.1 Three-Subsystem Hardware Partitioning

The reference S-Band full-duplex transceiver (2025–2110 MHz Uplink Rx / 2200–2290 MHz Downlink Tx) partitions hardware across three physical subsystems (**Figure 7.1**):

1. **Subsystem 1 — Digital Baseband FPGA:** Hosts `ccsds_sat_ttc_top`, the 16-bit LVDS Source-Synchronous Interface (`SSI`) bridge, the 4-wire SPI controller, and the < 500 ns hardware PA enable latch.
2. **Subsystem 2 — Agile Zero-IF SDR & Clock/PMIC Tree:** Performs direct quadrature RF conversion, closed-loop Digital Pre-Distortion (`DPD`), and phase-coherent 38.40 MHz clock distribution.
3. **Subsystem 3 — S-Band Full-Duplex RFFE:** Provides pre-PA spectral filtering, 4 W linear power amplification, reverse-power isolation, forward/reflected directional coupling, full-duplex diplexing, and ultra-low-noise receive amplification.

Board-level wiring and zone-by-zone schematic connections are detailed in Section 7.2, followed by component selection and vacuum-derated operating modes in Section 7.3.

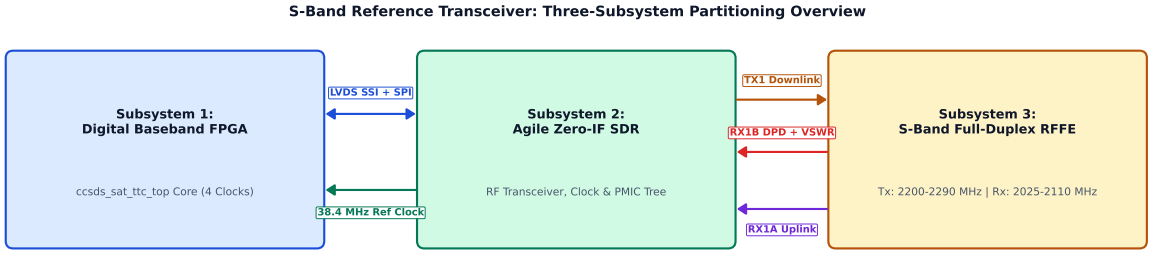


Figure 7.1: Three-subsystem hardware partitioning of the S-Band full-duplex reference transceiver.

7.2 Baseband-to-RF Interconnects & PA Protection

Figure 7.2 presents the inter-zone signal connectivity overview (top panel) followed by four focused zone sub-diagrams (**Figures 7.2a–d**), and **Table 7.1** lists the exact board-level pin routing and termination rules. Quantitative RF cascade placement proofs are derived in **Volume II, Section 7.1**; for board layout and bring-up, note three essential wiring rules in **Figure 7.2c**:

- **Upstream RF Isolator (#3) Before Bi-Directional Coupler (#2):** Output RF Isolator `#3` sits directly at the `SKY66294-11` PA output, while 20 dB Bi-Directional Coupler `#2` sits downstream between the Isolator and Diplexer `#4`. This ordering allows Coupler Port 4 to sense true antenna reflections before the isolator's 20 W termination load absorbs the reverse wave.
- **Sub-500 ns Hardware PA Cutoff (#8):** Coupler Port 4 feeds the **Analog Devices AD8362** True-RMS detector, whose output splits to (a) a **TI TLV4041R1** comparator ($V_{th} = 1.15$ V for VSWR $= 3.0:1$) that asynchronously clears the FPGA `PA_EN_LATCH` in < 500 ns, and (b) `ADRV9002 AUXADC_0` for continuous pass telemetry.
- **Closed-Loop DPD Observation (#7A):** Coupler Port 3 (forward coupled, -20 dB) passes through a 30 dB pad (`YAT-30A+`) into the `TCM2-33WX+` balun (`T203`) driving `ADRV9002 RX1B`.

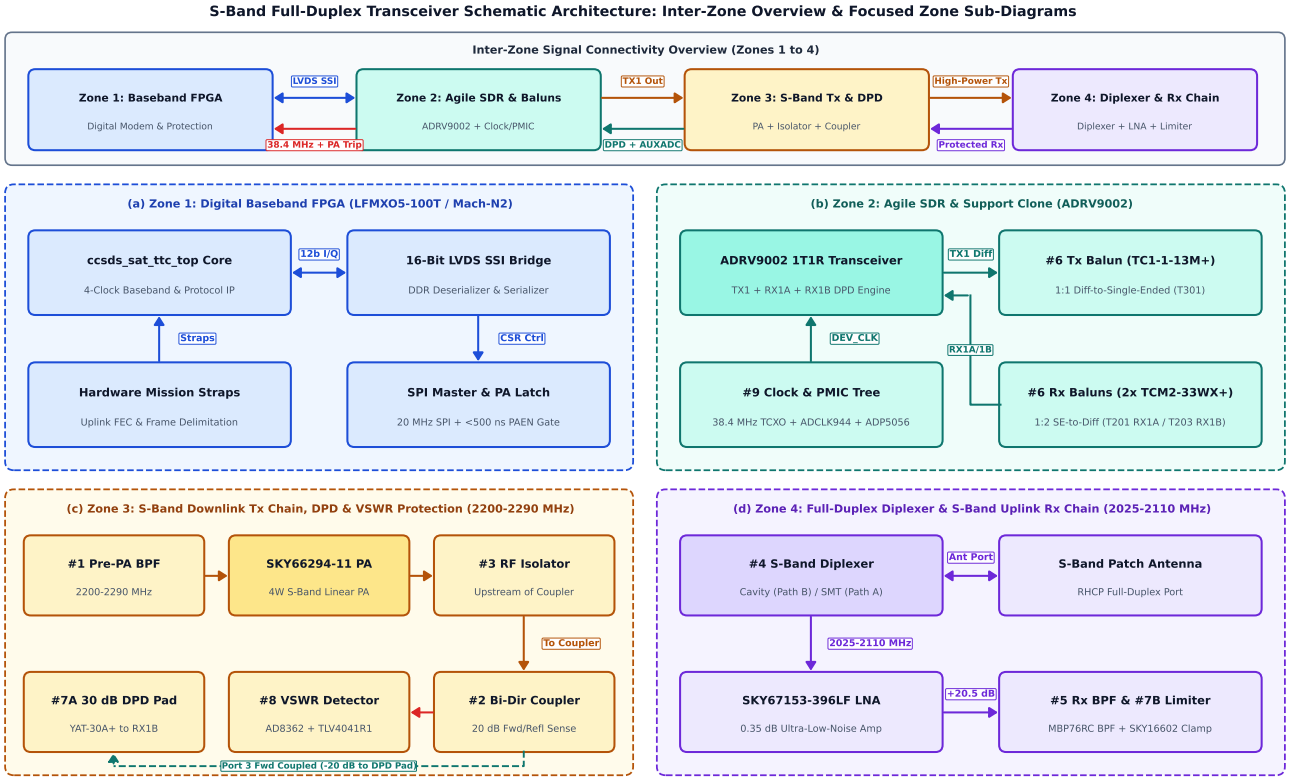


Figure 7.2: S-Band transceiver inter-zone connectivity overview and focused zone sub-diagrams (a–d).

Signal Group	Board Net Name(s)	I/O Standard	Pin Routing & Termination Requirement
Master Reference Clock	DEV_CLK_IN±, FPGA_REF_CLK±	LVPECL / LVDS (100 Ω)	Phase-synchronous 38.40 MHz fanout from ADCLK944BCPZ (U501) fed by GTXO-74V/JI TCXO (Y501).
Rx1A Baseband LVDS SSI	RX1_DCLK±, RX1_STROBE±, RX1_IDATA±, RX1_QDATA±	1.8 V LVDS (100 Ω)	DDR LVDS from ADRV9002 to FPGA Bank 2; deserializes 16-bit words and slices MSB [15:4] into rx_i/q_adc_in[11:0].
Tx1 Baseband LVDS SSI	TX1_DCLK±, RX1_STROBE±, TX1_IDATA±, TX1_QDATA±	1.8 V LVDS (100 Ω)	DDR LVDS from FPGA to ADRV9002; pads 12-bit phy_tx_12b as {phy_tx_12b, 4'b0000} into [15:0].
4-Wire Control SPI	SPI_CLK, SPI_CSB, SPI_MOSI, SPI_MISO	1.8 V LVCMOS	20 MHz SPI bus from FPGA SPI Master to ADRV9002 for boot init, 0.05 dB Tx attenuation, and AUXADC_0 readback.
Fast PA Enable & VSWR Trip	PA_FAULT_COMP_N → SKY_PAEN	1.8 V/3.3 V LVCMOS	Open-drain TLV4041R1 comparator output from AD8362; pulls SKY66294-11 PAEN low in < 500 ns on VSWR $> 3:1$.

Table 7.1: Board-level digital, clock, and PA protection interconnects between Zone 1, Zone 2, and Zone 3.

7.3 S-Band Bill of Materials & Operating Modes

The Flight PCB supports two passive filter assembly options: **Flight Path B** uses a 20 W-rated combline cavity diplexer (D0421552) operated at full +34.0 dBm (2.51 W) PA output, whereas **Flight Path A** uses a compact all-SMT 3.0 W-rated ceramic diplexer (CTS UPD215A) operated at a **NASA EEE-INST-002 / ECSS-Q-ST-30-11C 50% vacuum-derated +31.5 dBm (1.41 W) PA output**. For rapid reference, the bill of materials is split into the **Downlink Transmit & Protection Chain (Table 7.2)** and the **Diplexer & Uplink Receive Chain (Table 7.3)**.

Tx / Core Stage	Stage Function	Flight PCB Part (Path B / Path A)	PoC Bench Setup	Operating Level (Path B / Path A)
Core 1 + #9	FPGA + 38.4 MHz Clock	LFMXO5-100T + GTXO-74V + ADCLK944	LN2-MH-20-EVN + EVAL	SEL-immune FD-SOI FPGA + EVAL clock/PMIC tree.
Core 2 + #6 Tx	SDR + TX1 1 : 1 Balun	ADRV9002 + TC1-1-13M+ (T301)	On EVAL-ADRV9002	−1.7 dBm (Path B) / −3.4 dBm (Path A) out of T301.
Tx Stage 2 (#1)	Pre-PA Tx Bandpass	**B:** AE2250B10113 / **A:** BFCN-2275+	TB-270 (BFCN-2275+)	Strips Rx-band DAC noise; −2.5 dBm (B) / −5.0 dBm (A).
Tx Stage 3 (PA)	S-Band 4W Linear PA	Skyworks SKY66294-11 (+36.5 dB)	SKY66294-11-EVB	+34.0 dBm (2.51 W B) / +31.5 dBm (1.41 W A, derated).
Tx Stage 4 (#3)	Output RF Isolator	**B:** RFCI RFSL2848 / **A:** RFCR2848S	RFCI RFSL3818 (SMA)	Upstream of coupler; +33.80 dBm (B) / +31.25 dBm (A).
Tx Stage 5 (#2)	20 dB Bi-Dir Coupler	**B:** SYBD-20-272HP+ / **A:** BDCH-20-272+	ZABDC20-252H-S+	Downstream of isolator; +33.68 dBm (B) / +31.03 dBm (A).
DPD / VSWR (#7A/8)	DPD Pad & VSWR Trip	YAT-30A+ (T203) & AD8362 + TLV4041R1	SMA Pads + ADL5904	Port 3 → −16.2 dBm (RX1B); Port 4 →< 500 ns PAEN trip.

Table 7.2: Downlink transmit, DPD observation, and VSWR protection bill of materials (2200–2290 MHz).

Rx / Ant Stage	Stage Function	Flight PCB Part (Path B / Path A)	PoC Bench Setup	Operating Level (Path B / Path A)
Stage 6 / Rx 1 (#4)	Full-Duplex Diplexer	**B:** D0421552 (20 W) / **A:** UPD215A (3 W)	AE2067-2245DB5529	+32.9 dBm (1.95 W B) / +29.3 dBm (0.85 W A) at antenna.
Rx Stage 2 (LNA)	Ultra-Low-Noise Amp	Skyworks SKY67153-396LF (0.35 dB NF)	SKY67153-396EK2	+20.5 dB gain; sets 1.81–1.96 dB (B typ–max) NF.
Rx Stage 3 (#5)	Post-LNA Ceramic BPF	Sangshin MBP76RC2067S85A (1.2 dB IL)	Anatech AM2072B1583	65 dB typ (52 dB min) Tx rejection (> 119 dB total).
Rx Stage 4 (#7B)	PIN Diode RF Limiter	Skyworks SKY16602-632LF (0.5 dB IL)	SKY16602-632LF-EK1	+4 dBm flat-leakage spike clamp ahead of RX1A.
Rx Stage 5 (#6 Rx)	RX1A 1 : 2 Rx Balun	Mini-Circuits TCM2-33WX+ (T201)	On EVAL-ADRV9002	Locks in 1.81–1.96 dB (B) / 2.63–2.81 dB (A) system NF.

Table 7.3: Full-duplex diplexer and uplink receive bill of materials (2025–2110 MHz).

8 Verification Suite & Mission Sign-Off

To verify ccsds_sat_ttc_top in Xilinx Vivado 2022.2 xsim (or any IEEE 1800-2017 simulator), the repository provides automated Python and batch regression runners in scripts/. This chapter provides the execution commands, explains the relationship between open-loop waterfall stress sweeps (Stage 10) and rated-SNR full-chip sign-off (Stage 11), and summarizes the master verification matrix (**Table 8.1**).

8.1 Regression Execution & Verification Regimes

Run the complete regression suite or the 12-worker parallel 34,560-trial RF matrix from the repository root:

```
# 1. Run all unit, vertical-slice, multi-protocol, FEC/HCPD, and NIST/CCSDS golden reference checks:
python3 scripts/run_regression_suite.py

# 2. Launch the 12-worker parallel 5,760-config x 6-profile (34,560-trial) Full-Chip RF Matrix in Vivado xsim:
python3 scripts/run_stage11_matrix_parallel.py --workers 12 --mc-iters 6
```

Two methodological principles govern the verification campaigns (**Figure 8.1** and **Table 8.1**):

1. **Virtual-Fill Shortening in the 34,560-Trial Matrix (39,306,240 Baseband Samples):** Rather than simulating 34,560 unshortened 255-byte frames (~ 8 hours of RTL runtime), `tb_stage11_top_to_top_rf_link.sv` exercises CCSDS 131.0-B-5 implicit virtual fill $Q \in [185..205]$ with warm-locked carrier/timing loops across consecutive trials. Each bidirectional trial spans $1,137.3$ average I/Q samples ($39,306,240$ total samples), testing all finite-field, FEC, and SDLS state machines in 33.07 minutes.
2. **Open-Loop Breakdown Sweep (`Stage 10`) vs. Rated-SNR Sign-Off (`Stage 11`):** `Stage 10` (**Volume II, Figure 8.1**) sweeps down to $E_b/N_0 = 0.0$ dB to locate each codec's breakdown cliff. Conversely, `Stage 11` operates at rated mission SNR ($E_b/N_0 \geq 7.5$ dB) with deterministic error injection inside each codec's correction radius (1 bit in BCH, ≤ 6 bits in LDPC, ≤ 8 bytes in RS), proving that all 23,040 injected pre-FEC channel errors heal to **0 post-FEC bit errors**.

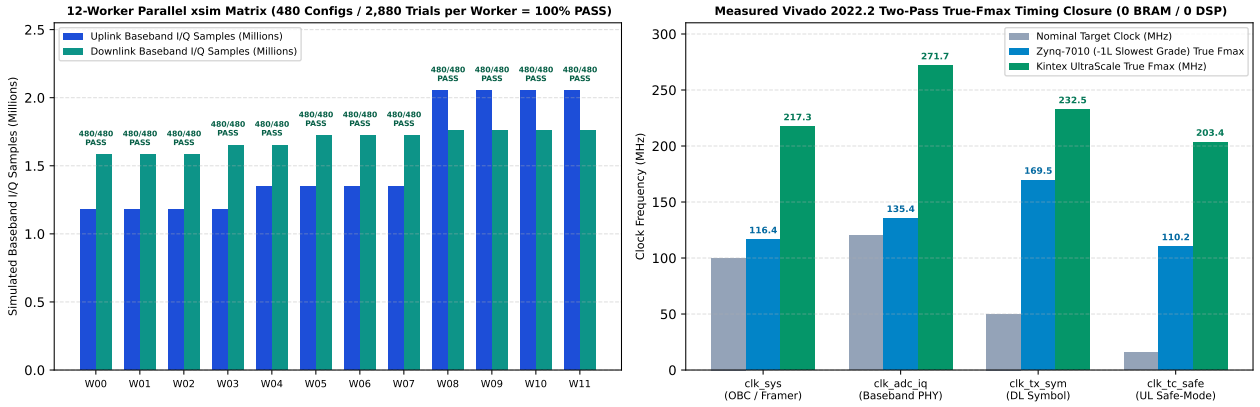


Figure 8.1: Twelve-worker parallel `xsim` sample volume across 34,560 trials (left) and measured Vivado 2022.2 two-pass $F_{\max}$ (right).

Verification Campaign / Suite	Channel Impairment & Configuration Scope	Measured Vivado 2022.2 `xsim` Sign-Off Result
`Stages 2-9`: Unit, Protocol, FEC & Golden Vector Suites	Tests CRC-16/32, $GF(2^8)$ dual-basis, Hsiao SECDED, all 5 framers/deframers, BCH/LDPC/RS/Viterbi, 3x TMR `FARM-1/HCPD`, and Python `cryptography` NIST `AES-128-GCM` vectors.	**PASS (100%):** Bit-exact match against golden reference vectors across all 8 foundational testbench stages.
`Stage 10`: Open-Loop Waterfall Sweep	23-point E_b/N_0 sweep (0.0–11.0 dB) across all uplink (BCH, LDPC) and downlink (Conv, RS, Concat, AR4JA, Turbo) codecs.	**PASS:** Waterfall knee positions match CCSDS theoretical bounds within ± 0.3 dB (**Volume II, Figure 8.1**).
`Stage 11`: 34,560-Trial Full-Chip RF Matrix	5,760 valid CCSDS configs $\times$ 6 channel profiles (`P0..P5`) with shortened virtual-fill frames and 23,040 injected pre-FEC bit errors at rated SNR.	**PASS (34,560 / 34,560):** Healed all 23,040 pre-FEC errors to **0 post-FEC bit errors** across 69,120 packets.
`Stage 12 CFG-1..3`: Gardner Timing, S-Curve Doppler & Ranging	28 multi-rate 4-SPS Gardner timing phases ($\tau \in \{0..3\}$), LEO S-curve Doppler (± 512 NCO offset, 4 turnaround ratios), and `T4B/T2B` PN ranging.	**PASS:** Locked all 28 timing phases, 4 turnaround ratios, and $1,009,470$-chip PN ranging with **0 bit errors**.
`Stage 12 CFG-4..6`: 5-Protocol CADU, Deep-Fade Replay & `HCPD`	Runtime switching across all 5 protocols, mid-pass Rician deep-fade frame erasure with automatic COP-1 Go-Back-N replay, and `VCID=63 HCPD` reset.	**PASS:** Restored `V(R)=3` after deep fade with **0 lost/duplicate packets** and verified `hcpd_obc_reset_pulse`.

Table 8.1: Master verification sign-off summary for `ccsds_sat_ttc_top (v2.40.3)`.

References

[1] CCSDS 133.0-B-2, *Space Packet Protocol*, Blue Book, Issue 2, June 2020.

[2] CCSDS 132.0-B-3, *TM Space Data Link Protocol*, Blue Book, Issue 3, Oct 2021.

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[4] CCSDS 232.1-B-2, *Communications Operation Procedure-1 (COP-1)*, Blue Book, Issue 2, Sept 2010.

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[7] CCSDS 211.0-B-6, *Proximity-1 Space Link Protocol—Data Link Layer*, Blue Book, Issue 6, July 2020.

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[10] CCSDS 231.1-O-1, *Dedicated Short-Block Uplink LDPC Codes*, Orange Book, Issue 1, April 2015.

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[13] CCSDS 401.0-B-32, *Radio Frequency and Modulation Systems—Part 1: Earth Stations and Spacecraft*, Blue Book, Oct 2021.

[14] CCSDS 414.1-B-3, *Pseudo-Noise (PN) Ranging Systems*, Blue Book, Issue 3, July 2022.